EtherCAT User Guide

EzFamily Series
EtherCAT

AJINEXTEK AXT

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Content

CONTENT 3

1. EZSOFTWARE INSTALLATION AND UNINSTALLATION 6

1.1 BIOS Setup 6

1.1.1 Disable Hyper-Threading 6
1.1.2 Disable CPU Sleep Status switching (C State) 7
1.1.3 Disable Intel SpeedStep Technology (EIST) 8
1.1.4 Disable Intel Turbo Boost Technology 9

1.2 EzSoftwareUC Installation 10

1.2.1 Mount PCIe-RxxIF-ECAT Board 10
1.2.2 Mount USB Type EtherCAT Master Key 11
1.2.3 EzSoftwareUC Package Installation 13
1.2.4 Check If EzSoftwareUC is Installed Normally 18

1.3 EzSoftwareUC Uninstallation 20

1.3.1 Exit EzManager 20
1.3.2 Remove EzSoftwareUC 22

2. CONSTRUCT OPERATING ENVIRONMENT FOR ETHERCAT MASTER MANUALLY 23

2.1 Register network port to RTOS 23

2.1.1 List of network cards supported by AJINEXTEK EtherCAT Master 23
2.1.2 Register Network Port for RTOS 26

2.2 Allocate CPU Core and Memory to RTOS 29

3. EZMANAGER 31

3.1 Layout 31

3.1.1 Tray Menu 31
3.1.2 Realtime Layout 33

3.1.3 Online Layout 36
3.1.4 Devices Layout 38
3.1.5 Update Layout 39
3.1.6 About Layout 42

3.2 Menu 43

3.2.1 File Menu 43
3.2.2 Option Menu 44
3.2.3 View Menu 45
3.2.4 Help Menu 47

4. ERROR CORRECTION 48

4.1 If EtherCAT Master is not running 48

4.1.1 If EtherCAT Master Key is not installed 48
4.1.2 If License Manager Program (Codemeter) is not installed correctly 49
4.1.3 If EtherCAT Master Key Driver is not installed 50
4.1.4 If EtherCAT Master Key is not recognized 51
4.1.5 If CPU core for EtherCAT Master is not allocated 52
4.1.6 If return code 0x106B is output and RTOS is not executed 53

4.2 If EtherCAT Slave is not detected in EzConfig 54

4.2.1 If Network Cable is not connected correctly 54
4.2.2 If slave is not detected 55
4.2.3 If ESI file does not exist for slave 56
4.2.4 Save EtherCAT Master log 57

5. APPENDIX 58

5.1 Term Explanation 58

Revision History

Version Info	Date & Version	Comment
Rev. 1.0 issue 1.0	2015.12.02	First Release
Rev. 1.1 issue 1.1	2016.04.14	Add Network port distinction and BIOS setup
Rev. 1.2 issue 1.2	2016.06.08	Update contents according to UI change
Rev. 1.3 issue 1.3	2016.07.11	Update BIOS setup and note screenshots
Rev. 1.4 issue 1.4	2016.11.17	Add USB Type Master Key Add Error Correction
Rev. 1.5 issue 1.5	2017.02.14	Add Save EtherCAT Master log

1. EzSoftware Installation and Uninstallation

1.1 BIOS Setup



Be careful when making changes in the BIOS Setup Utility. If you make a mistake, your PC may not work as expected.

Our Company's EtherCAT product is based on Real-Time OS(RTOS). Since the RTOS uses a separate physical CPU core, it is affected by the real-time control performance of the RTOS, depending on how the BIOS are set up. Before installing EzSoftware package, let's look at how to change the system's BIOS settings.

Caution) There may be no setting items depending on whether the name of the BIOS is different or whether the main board supports the corresponding function.

1.1.1 Disable Hyper-Threading

Hyper-Threading or Simultaneous multithreading is an option that supports multithreading. If you use this option, the CPU core allocated by the RTOS cannot be used by itself. Due to this reason **this option must be disabled**. Please refer to the BIOS access procedure for each mainboard manufacturer.

Manufacturer	Option Location
ADLINK	[Advanced] -> [CPU Configuration] -> [Hyper-threading]
ADVANTECH	[Advanced] -> [CPU Configuration] -> [Hyper-threading]
ASRock	[Advanced] -> [CPU Configuration] -> [Intel Hyper Threading Technology]
ASUS	[Advanced] -> [CPU Configuration] -> [Hyper-threading]
GIGABYTE	[M.I.T.] -> [Hyper-Threading Technology]
SUPERMICRO	None

1.1.2 Disable CPU Sleep Status switching (C State)

CPU C State is a function that turns non-working cores into the Sleep state in order to reduce power consumption according to the load applied to the CPU core. If this option is used, the real-time control performance of the RTOS will be reduced because CPU clock fluctuation increases. Due to this reason **this option must be disabled**. Please refer to the BIOS access procedure for each mainboard manufacturer.

Manufacturer	Option Location
ADLINK	[Advanced] -> [CPU Configuration] -> [C1E Function] [Advanced] -> [CPU Configuration] -> [CPU C3 Report] [Advanced] -> [CPU Configuration] -> [CPU C6 Report] [Advanced] -> [CPU Configuration] -> [CPU C7 Report]
ADVANTECH	[Advanced] -> [CPU PPM Configuration] -> [CPU C3 Report] [Advanced] -> [CPU PPM Configuration] -> [CPU C6 Report] [Advanced] -> [CPU PPM Configuration] -> [CPU C7 Report]
ASRock	[Advanced] -> [CPU Configuration] -> [CPU C States Support] [Advanced] -> [CPU Configuration] -> [CPU C3 State Support] [Advanced] -> [CPU Configuration] -> [CPU C6 State Support] [Advanced] -> [CPU Configuration] -> [CPU C7 State Support]
ASUS	[Advanced] -> [CPU Configuration] -> [CPU C States Support] [Advanced] -> [CPU Configuration] -> [CPU C3 State Support] [Advanced] -> [CPU Configuration] -> [CPU C6 State Support] [Advanced] -> [CPU Configuration] -> [CPU C7 State Support]
GIGABYTE	[M.I.T.] -> [CPU Enhanced Halt(C1E)] [M.I.T.] -> [C3 State Support] [M.I.T.] -> [C6/C7 State Support] [M.I.T.] -> [C8 State Support]
SUPERMICRO	[Advanced] -> [CPU Configuration] -> [Power Technology]

1.1.3 Disable Intel SpeedStep Technology (EIST)

Intel SpeedStep Technology and EIST functions are designed to speed up work by changing the CPU clock according to workload. If this option is used, the real-time control performance of the RTOS will be reduced because CPU clock fluctuation increases. Due to this reason **this option must be disabled**. Please refer to the BIOS access procedure for each mainboard manufacturer.

Manufacturer	Option Location
ADLINK	[Advanced] -> [CPU Configuration] -> [EIST]
ADVANTECH	[Advanced] -> [CPU PPM Configuration] -> [EIST]
ASRock	[Advanced] -> [CPU Configuration] -> [Intel SpeedStep Technology]
ASUS	[Advanced] -> [CPU Configuration] -> [EIST]
GIGABYTE	[M.I.T.] -> [CPU EIST Function]
SUPERMICRO	None

1.1.4 Disable Intel Turbo Boost Technology

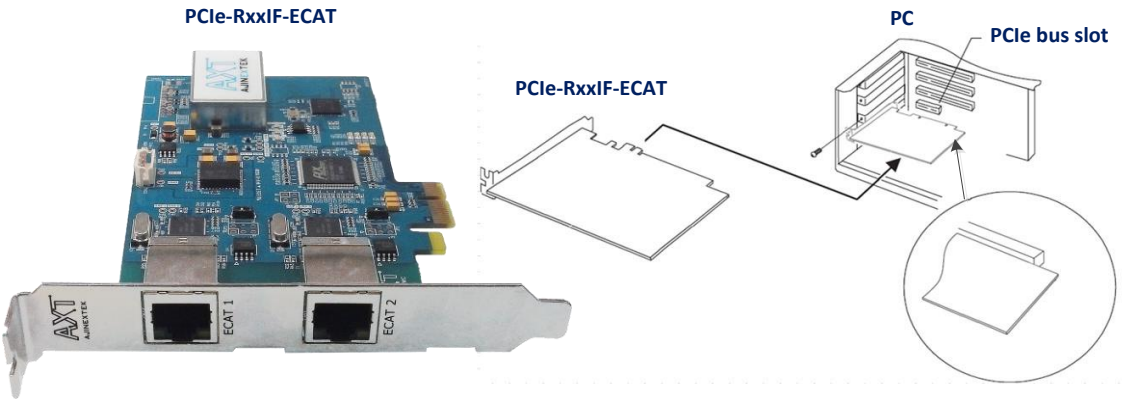
Intel Turbo Boost Technology function allows the CPU clock to be automatically raised above the displayed reference clock for faster processing when the processor is below the specified power consumption, temperature, or current. If this option is used, the real-time control performance of the RTOS will be reduced because CPU clock fluctuation increases. Due to this reason **this option must be disabled**. Please refer to the BIOS access procedure for each mainboard manufacturer.

Manufacturer	Option Location
ADLINK	None
ADVANTECH	None
ASRock	[Advanced] -> [CPU Configuration] -> [Intel Turbo Boost Technology]
ASUS	[Advanced] -> [CPU Configuration] -> [Turbo Mode]
GIGABYTE	[M.I.T.] -> [Intel Turbo Boost Technology]
SUPERMICRO	[Advanced] -> [CPU Configuration] -> [Turbo Boost Technology] -> [Turbo Mode]

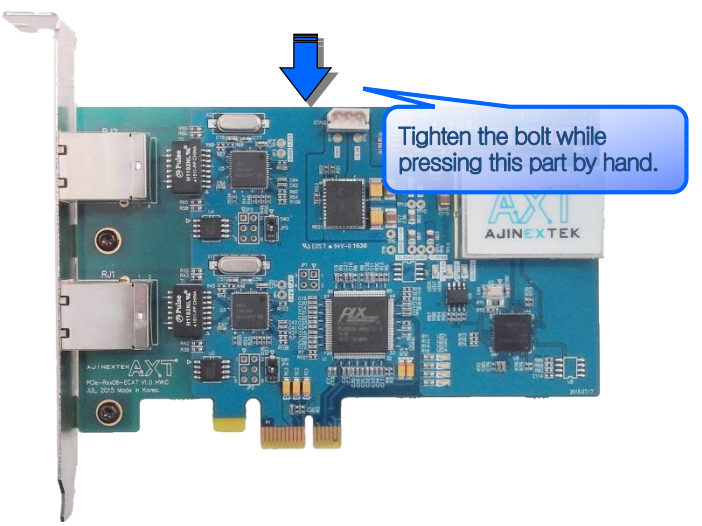
1.2 EzSoftwareUC Installation

1.2.1 Mount PCIe-RxxIF-ECAT Board

1. Before applying power to the system to install EzSoftwareUC, mount PCIe-RxxIF-ECAT board purchased for EtherCAT control into PCI Express slot.



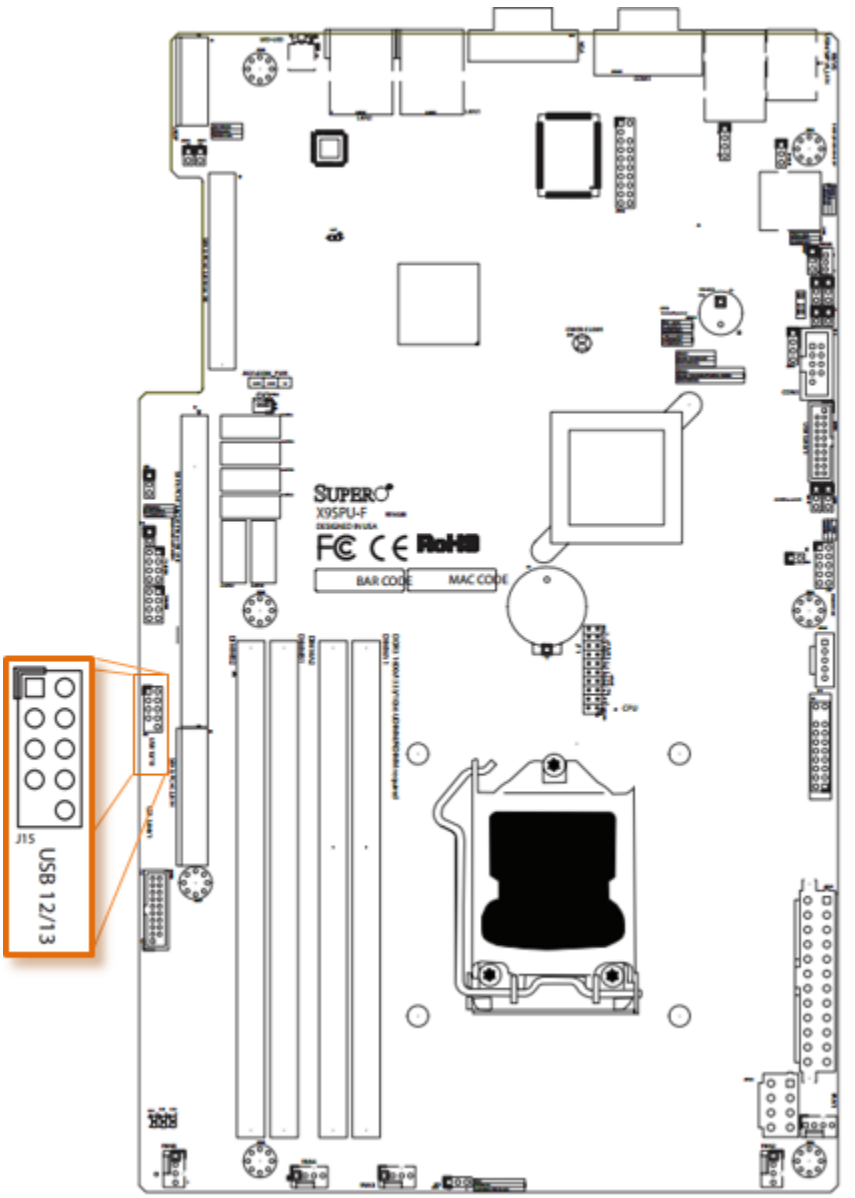
2. When mounting the PCIe-RxxIF-ECAT board in the system, press the upper part of the board with your hand so that the edge of the board is completely in contact with the bottom of the PCI Express slot, and tighten the bolts to secure the board. If the edge of the board is mounted wrongly, the board may not be recognized by the control system.



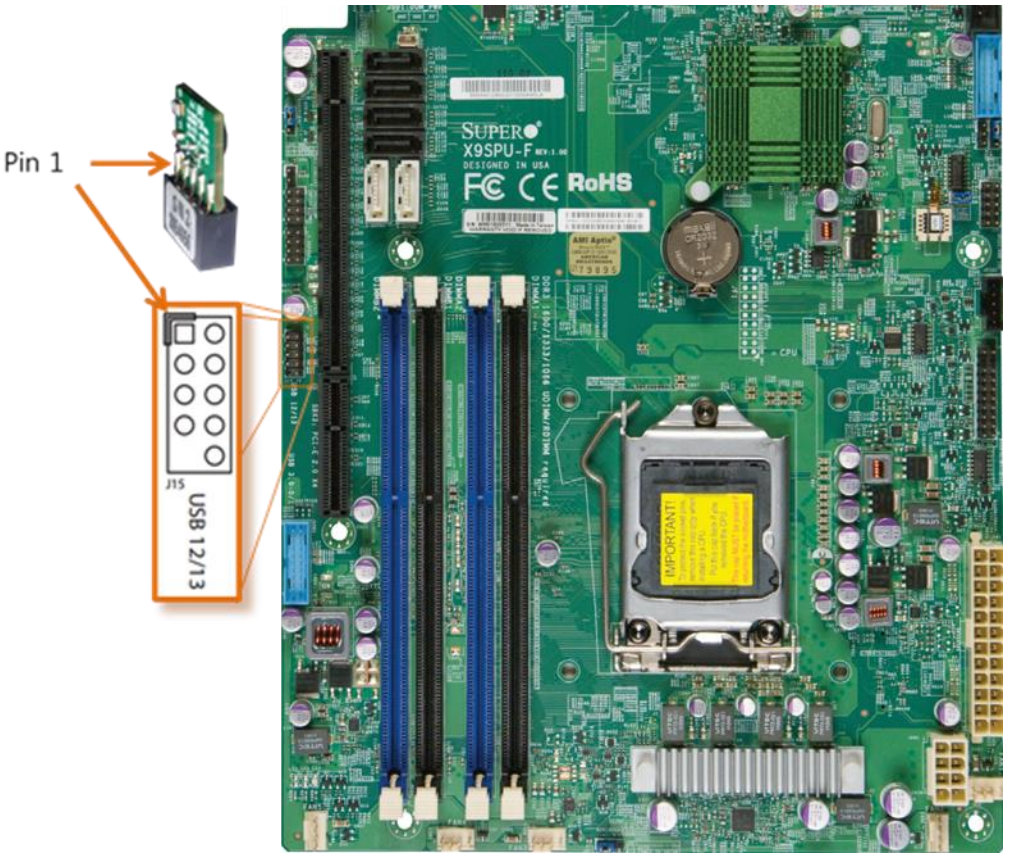
1.2.2 Mount USB Type EtherCAT Master Key

In order for EtherCAT Master to operate properly, the system must be equipped with H/W EtherCAT Master Key. When using the Ethernet port installed in the control system without using the PCIe-RxxIF-ECAT board for EtherCAT communication, install the separate EtherCAT Master Key as follows.

1. Check pin 1 on the 2x5 pin type USB connector on the CPU board. Typically, pin 1 uses a quadrangular silk or marks with dark silk at the corner.

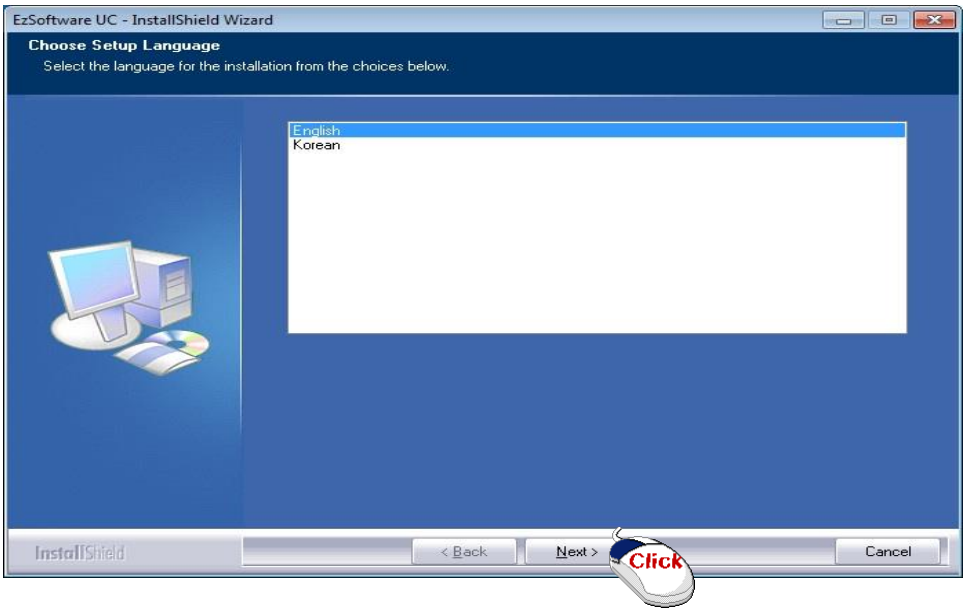


2. Install the CPU board with the USB connector pin 1 and the EtherCAT Master Key pin 1 aligned. If the wrong direction is installed, it will not cause a problem with the product. However, the license is not recognized so the EtherCAT Master cannot be used.

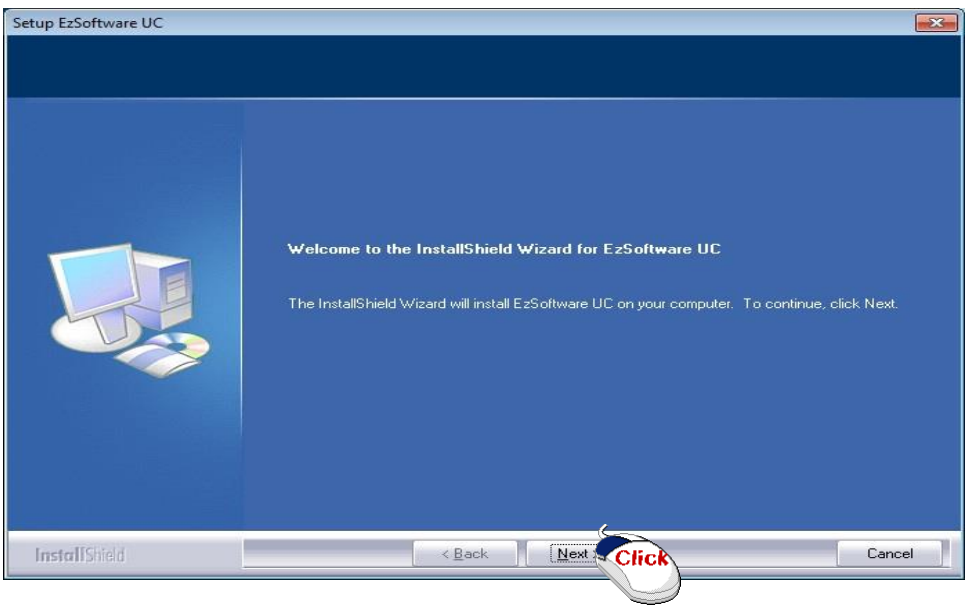


1.2.3 EzSoftwareUC Package Installation

1. Double-click the EzSoftwareUC installation file to start the installation. First, select the language you want to use for the installation and click the "Next" button to continue.



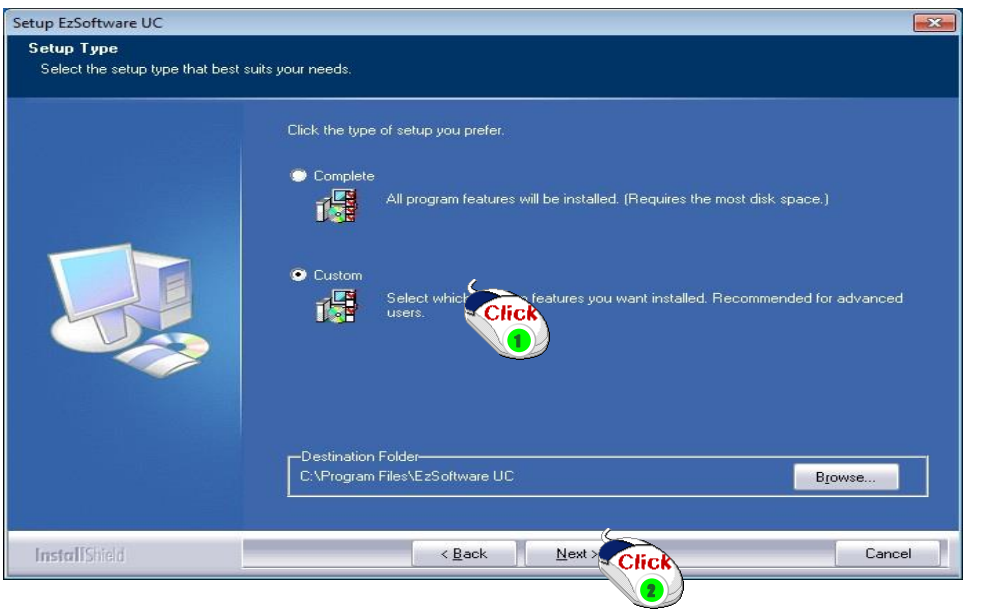
2. Confirm the installation of EzSoftwareUC. Click "Cancel" to cancel the installation or click the "Next" button to continue.

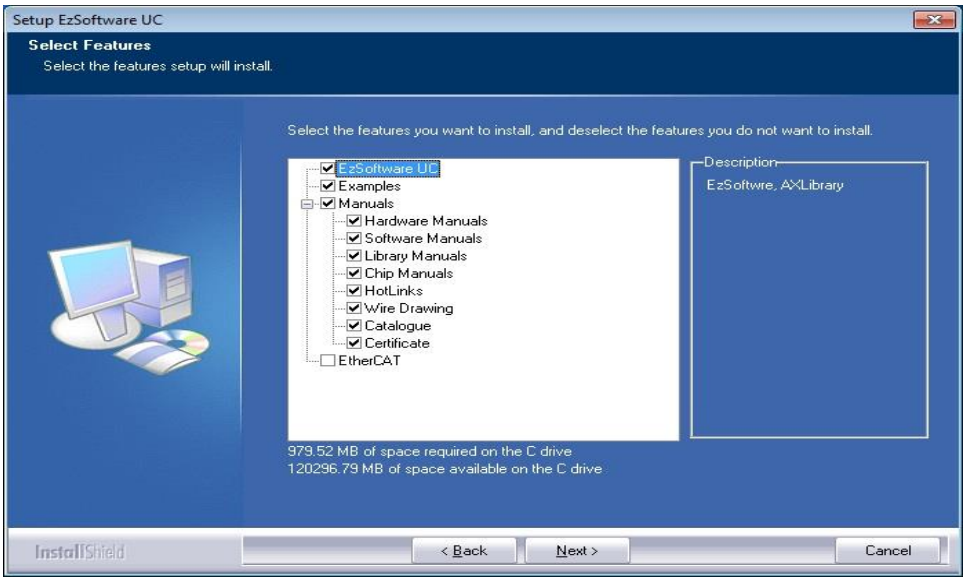


3. Confirm the license agreement of EzSoftwareUC. You must agree to all the terms of the agreement to proceed. If you agree, click "Next" button to continue the installation.



4. Select the installation type of EzSoftwareUC. Selecting "Complete" will install all components included in the package. "Custom" allows you to proceed with the installation by selecting only the required components of the entire installation.





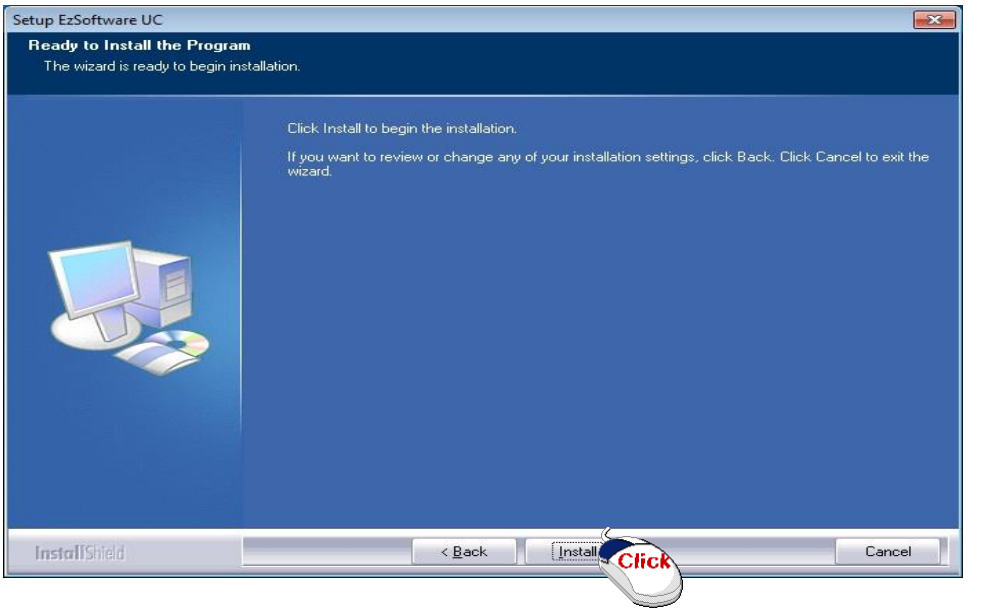
Caution) If you do not select the EtherCAT item, the EtherCAT controller will not be available for that system because the EtherCAT Master running environment is not installed.

5. If you want to install EzSoftwareUC with minimum specifications requirements for EtherCAT, select only EtherCAT item.

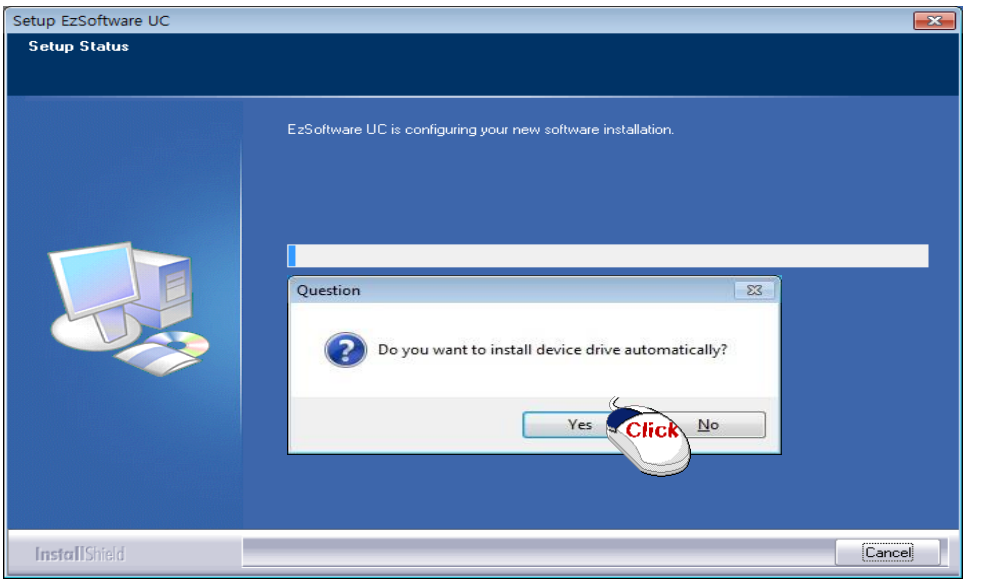
[Automatically installed and changed when selecting EtherCAT item]

Item	Description
USB Driver	USB driver for checking EtherCAT license installed on PCIe-RxxIF-ECAT board
CodeMeter	Applications for managing EtherCAT licenses for use with CodeMeter
CPU Core Allocation	Automatically allocate CPU to be used for RTOS
RTOS operating environment installation	Install RTOS operating environment (RTOS service, RTOS Driver, Memory allocation, etc.)
ESI File installation	Controllable EtherCAT Slave Information File installation
EzManager installation	Install EzManager application to run and manage RTOS

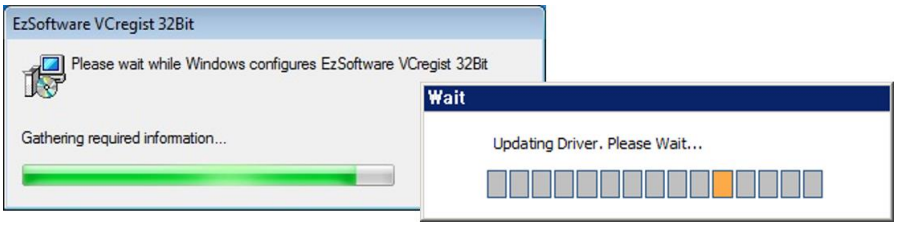
6. Confirm to proceed with the installation. Click the "Install" button to proceed with the installation of EzSoftwareUC.



7. After the installation files are copied, the device driver registration for the hardware devices proceeds. If a message box asking whether to proceed with registration automatically appears, click "Yes" button to proceed with driver installation automatically. When reinstalling the package, you may not automatically register the device driver as required.



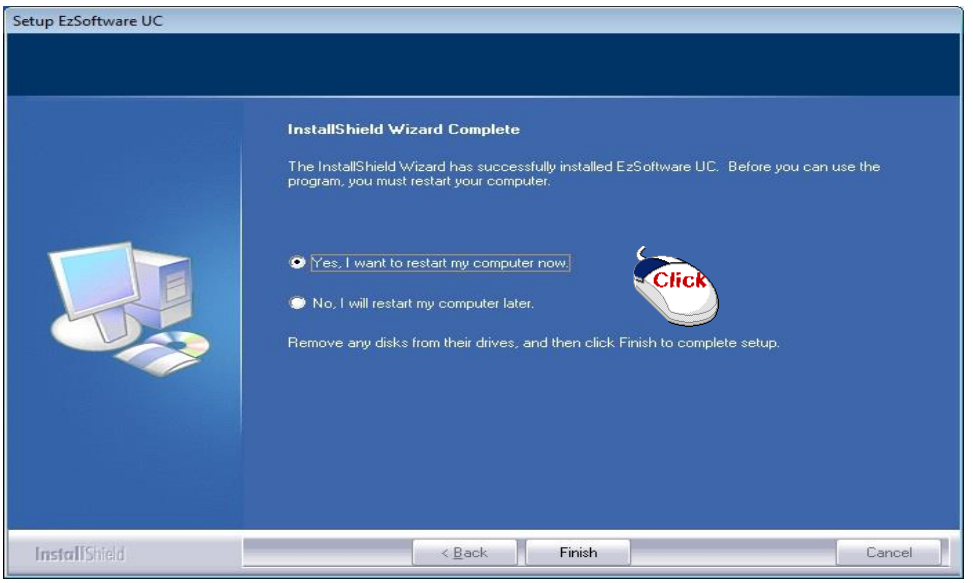
8. When the device driver installation for the hardware devices provided by us is completed, the redistribution package is installed so that the software applications included in the EzSoftwareUC package can be normally executed, and finally the development environment of the EtherCAT product setting proceed.



Caution) If Windows security-related message appears during device driver installation, be sure to click "Install" to continue the installation.



9. When the installation is completed successfully, the installation completion screen will appear as shown in the following screen. If system reboot is possible, select "Yes, I want to restart my computer now." to start rebooting automatically. If you are copying large files or other important applications are running (or saving), select "No, I will restart my computer later" and click "Finish" button to complete the installation without rebooting the system.



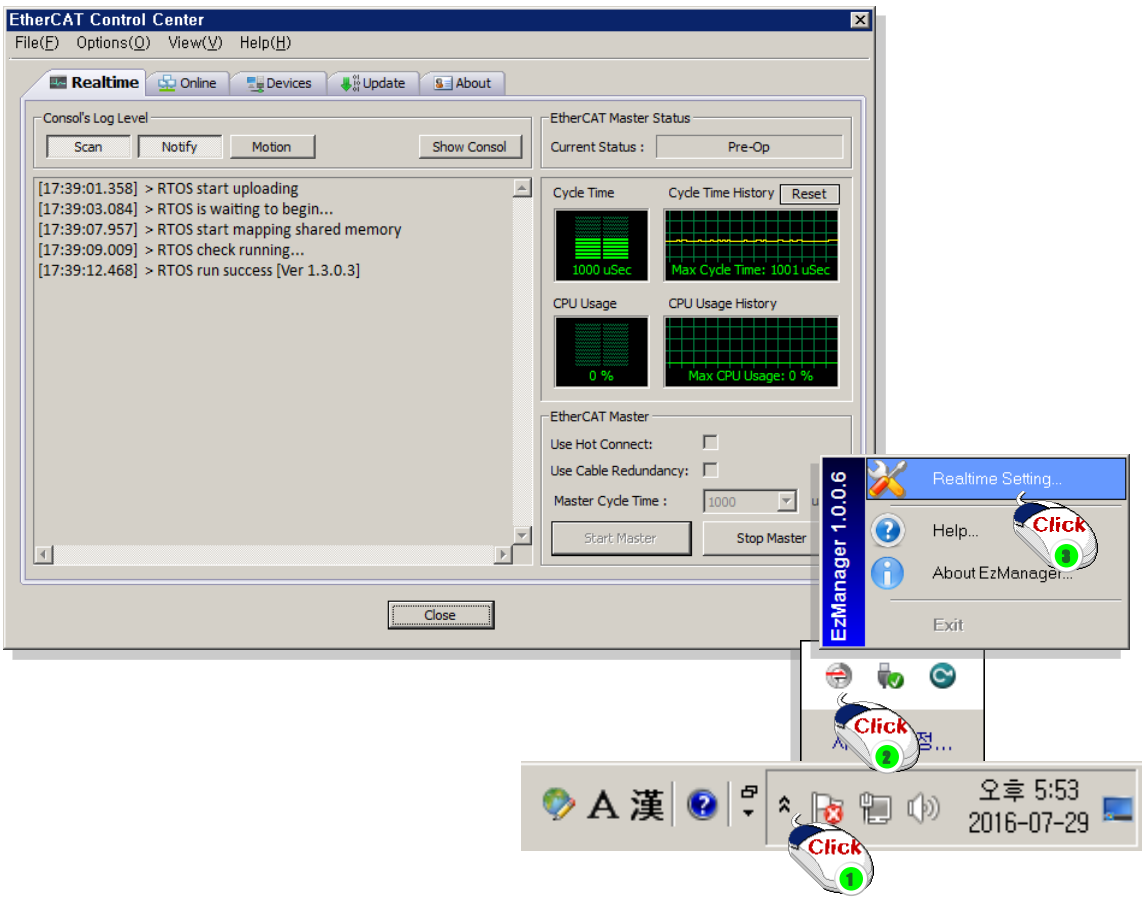
1.2.4 Check If EzSoftwareUC is Installed Normally

1. After installing EzSoftwareUC (if you installed the whole installation or EtherCAT item in step 4 of the manual 1.2.3), you will need to reboot the system to build the operating environment of EtherCAT Master. To restart the system automatically click the "Restart Now" button.

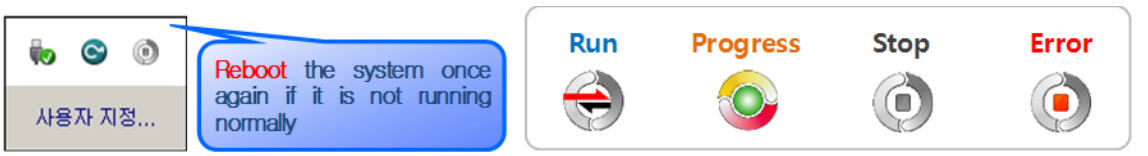


Caution) If the system is not restarted, EtherCAT Master may not operate normally.

2. The EtherCAT Master starts automatically after rebooting. To check whether it is running properly, click the EzManager tray icon at the bottom right of the Windows task bar and select "Realtime Setting ... " to launch the EtherCAT Control Center.



3. If "RTOS run success [Master x.x.x.x]" is displayed in the log window of the "Realtime" tab, the EtherCAT Master is running normally. If the installation environment of the EtherCAT Master is not completed normally (such as allocating memory to the RTOS), a message may ask to restart the system. These reboot requests are only generated when you first build the EtherCAT control environment after installing Windows on your system.



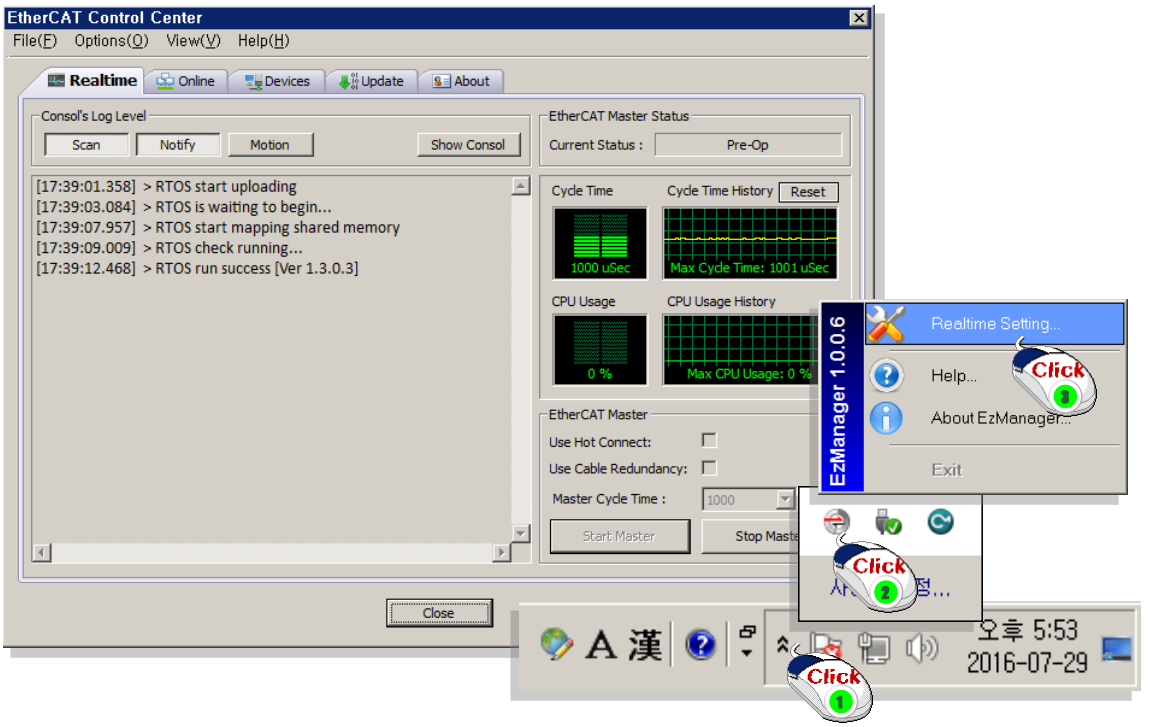
Caution) If EtherCAT Master doesn't operate normally after rebooting more than 3 times after installing EzSoftwareUC, please refer to the "Error Correction" in the manual.

1.3 EzSoftwareUC Uninstallation

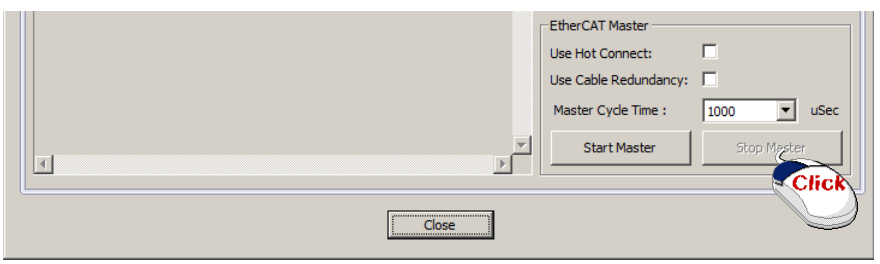
1.3.1 Exit EzManager

EzManager is a Windows application that manages the EtherCAT Master. You can't remove EzSoftwareUC while the EtherCAT Master is running, so you must stop and remove the EtherCAT Master via EzManager.

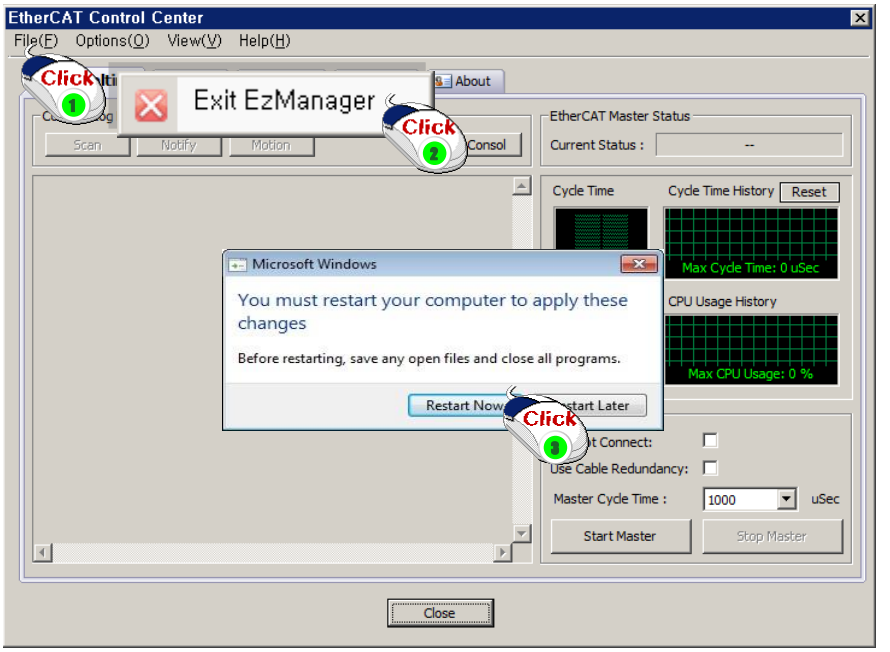
1. Click the EzManager program icon at the bottom right of the Windows task bar and select "Realtime Setting ..." from the menu to launch the EtherCAT Control Center.



2. In the "Realtime" tab, click "Stop Master" to stop EtherCAT Master. If EtherCAT Master is stopped, EtherCAT can't be controlled. Therefore, use it only when removing EzSoftwareUC or re-running EtherCAT Master.

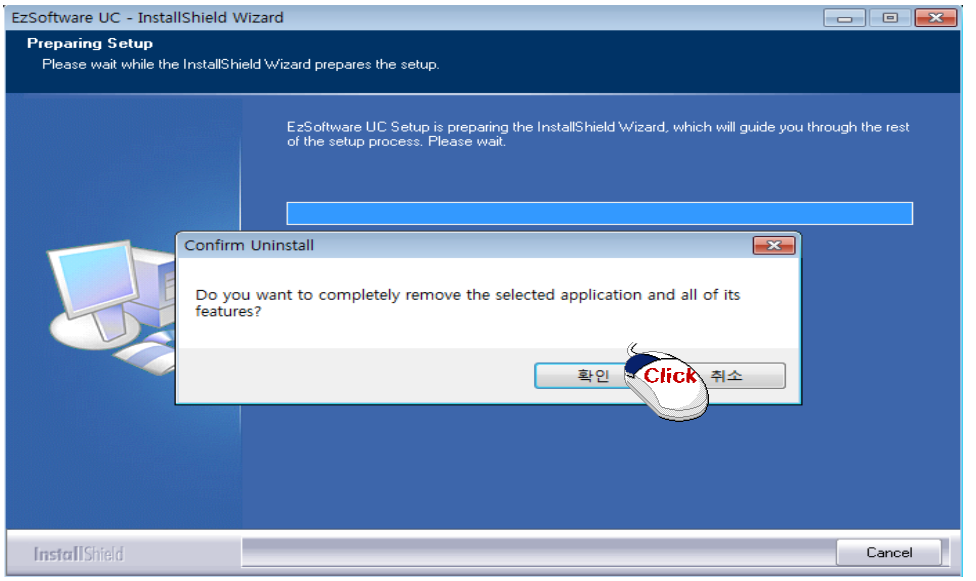


3. When the EtherCAT Master is shut down, select "File (F) >> Exit EzManager" menu at the top of EtherCAT Control Center to exit the program. The menu is disabled while the EtherCAT Master is running.

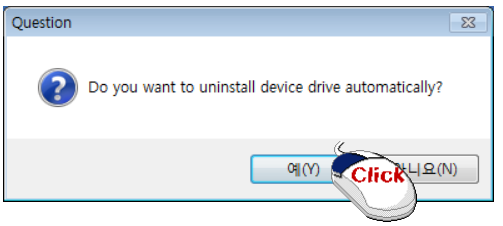


1.3.2 Remove EzSoftwareUC

1. There are 3 ways to remove EzSoftwareUC. First, click Add or Remove Programs in Control Panel of Windows and click Change / Remove (C) of EzSoftwareUC. Second, select "Remove EzSoftwareUC -> EzSoftwareUC" in Windows Start Menu. Finally, run "Setup.exe" file from the installation CD provided when you purchased the product.



2. The installed components are removed after the installed device driver to control the hardware is removed. If the message "Do you want to uninstall the device driver automatically?" appears during the uninstallation process, select "Yes (Y)" to continue uninstallation.



Caution) If the device driver is reinstalled with the same version after removing EzSoftwareUC, it doesn't need to remove device driver in this step.

3. When the EzSoftwareUC removal is completed, reboot the system so Windows can use the resources that allocated for running EtherCAT Master.

2. Construct Operating Environment for EtherCAT Master Manually

2.1 Register network port to RTOS

EtherCAT communication is using Ethernet port to control slaves. Since the EtherCAT Master provided by our company is operated in the RTOS environment, the control system must be equipped with an Ethernet port that can be controlled by the RTOS, and the network adapter driver must be registered for RTOS support.

2.1.1 List of network cards supported by AJINEXTEK EtherCAT Master

Link Layer Name	Controller	Device ID(Hex)
Emllll8255x Intel Pro/100	82551QM	0x1059
	82555VE2	0x27DC
	82557	0x1229
	82557ER	0x1209
	82559ER	0x2449
	82562	0x1039
	82801DB	0x103A
	82801EB	0x1050
	Pro/100/M	0x1229
	Pro/100/S	0x1229
	Pro/100/VE	0x1092
emlll8254x Intel Pro/1000	82540EM	0x100E
	82541EI	0x1013
	82541ER	0x1078
	82541GI	0x1076, 0x1077
	82541PI	0x107C
	82545GM	0x1026
	82546EB	0x1010
	82546GB	0x1079
	82547EI	0x1075
	82547GI	0x1019
	82566DM	0x104A
	82566L	0x10BD
	82566MC	0x104D

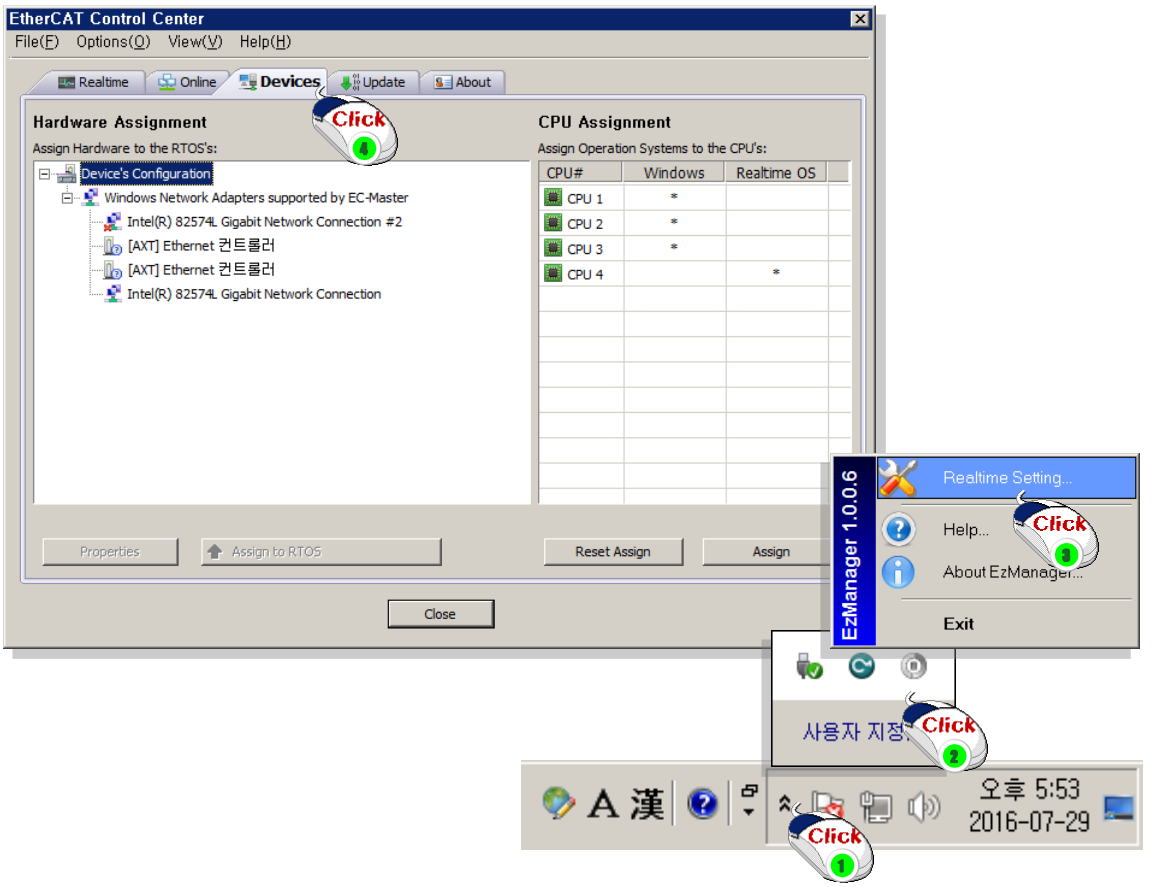
	82567V	0x10CE , 0x1501
	82567LM	0x10DE, 0x10F5
	82571GB	0x10A4, 0x10BC
	82572GI	0x10B9
	82572PI	0x107D
	82573	0x108C
	82573E	0x108B
	82573L	0x109A
	82574(L)	0x10D3
	82575	0x10A7
	82577LM	0x10EA
	82577LC	0x10EB
	82576	0x10C9
	82576 ET2	0x1526
	82578DM	0x10EF
	82578DC	0x10F0
Link Layer Name	**Controller**	**Device ID(Hex)**
emllI8254x Intel Pro/1000	82579LM	0x1502
	82579V	0x1503
	82580	0x150E
	82580 QF	0x1527
	82583V	0x150C
	N1E5132	0x105E
	I350	0x1521
	I210	0x1533
	I210 CFL	0x157B
	I211AT	0x1539
	I217LM	0x153A
	I217V	0x153B
	I218LM	0x155A
	I218V	0x1559
emllRTL8139 Realtek RTL8139	8139D	0x8139
	D-Link 8139D	0x1300
emllRTL8169 Realtek Gigabit NIC (PCIe, PCI)	RTL8110	0x8169
	RTL8111	0x8168
	RTL8168	0x8168
	RTL8169	0x8169
	D-Link RTL8169	0x4300
	RTL8169SC	0x8167
	RTL8103	0x8136
emllCPSW	Texas Instruments Sitara on board	

emllETSEC	Freescale PowerPC TSEC/eTSEC	
	eTSEC v1 and v2	
emllFslFec	Freescale FEC and ENET controller	
emllR6040	RDC R6040	

2.1.2 Register Network Port for RTOS

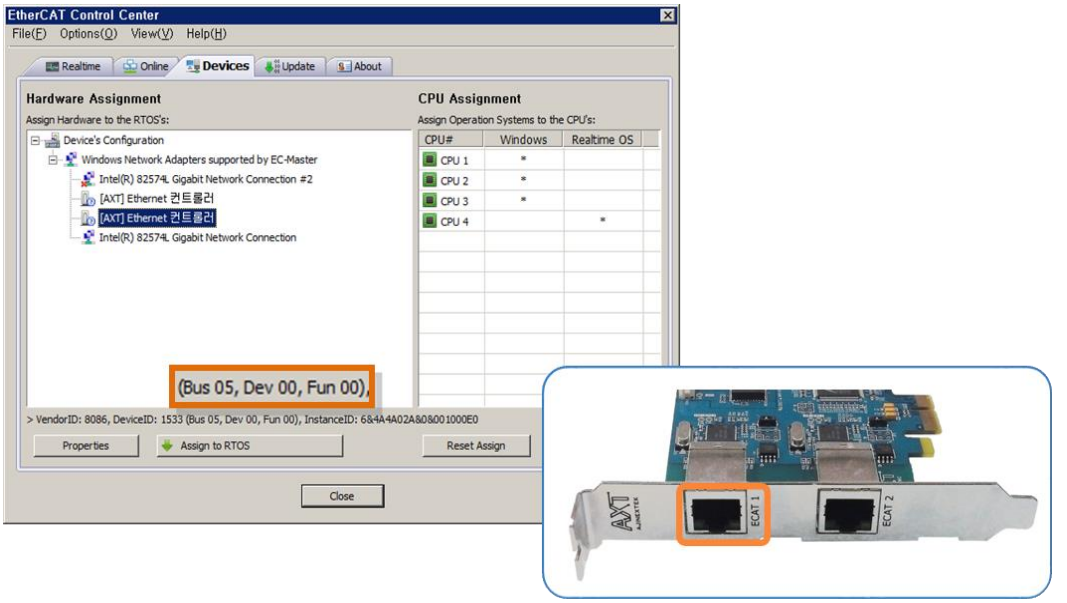
The network adapter driver will be automatically registered to the RTOS during the installation if the PCIe-RxxIF-ECAT board is installed in the control system before installing the EzSoftwareUC package. However, if the board is installed in the control system after installation of the package or if you use the above-mentioned separate Ethernet card, the user must manually register the network adapter in the RTOS.

1. Click the EzManager tray icon in the bottom right corner of the Windows task bar and select "Realtime Setting...". Select the "Devices" tab in the EtherCAT Control Center to start the driver registration.

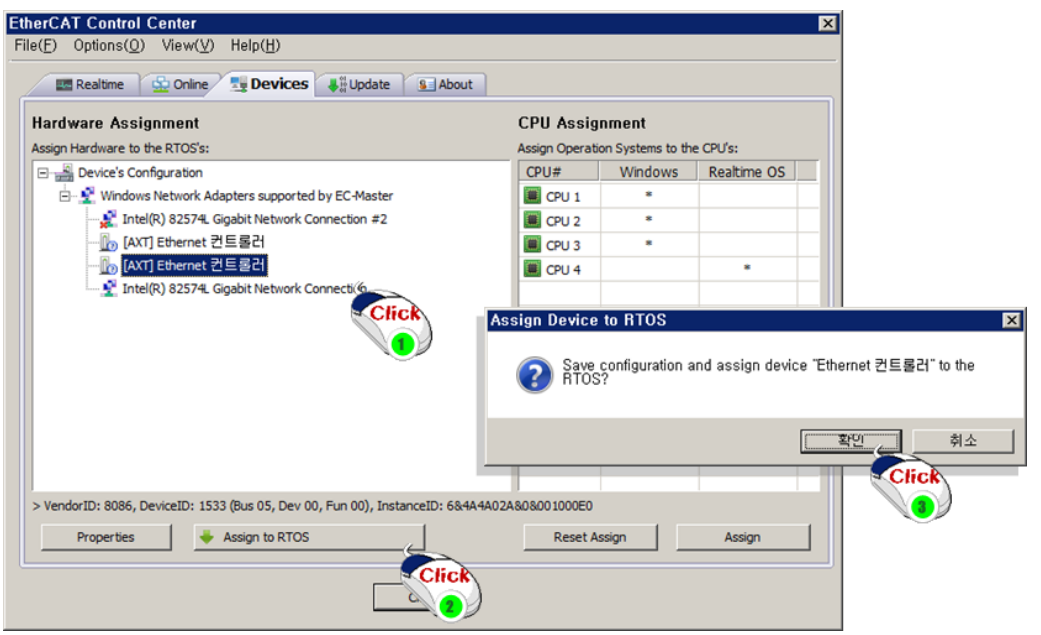


2. Select the network ports on the Assign Hardware to the RTOS's tree to register in the RTOS. The port with the "[AXT]" mark means the port attached to the PCIe-RxxIF-ECAT card. Among the two [AXT] ports, the first registered port is designated as the EtherCAT Main port.

3. Among the two ports on the PCIe-RxxIF-ECAT card, the lower bus number is ECAT1. Including the bus number, detailed information appears when clicking the each port on the tree bar.

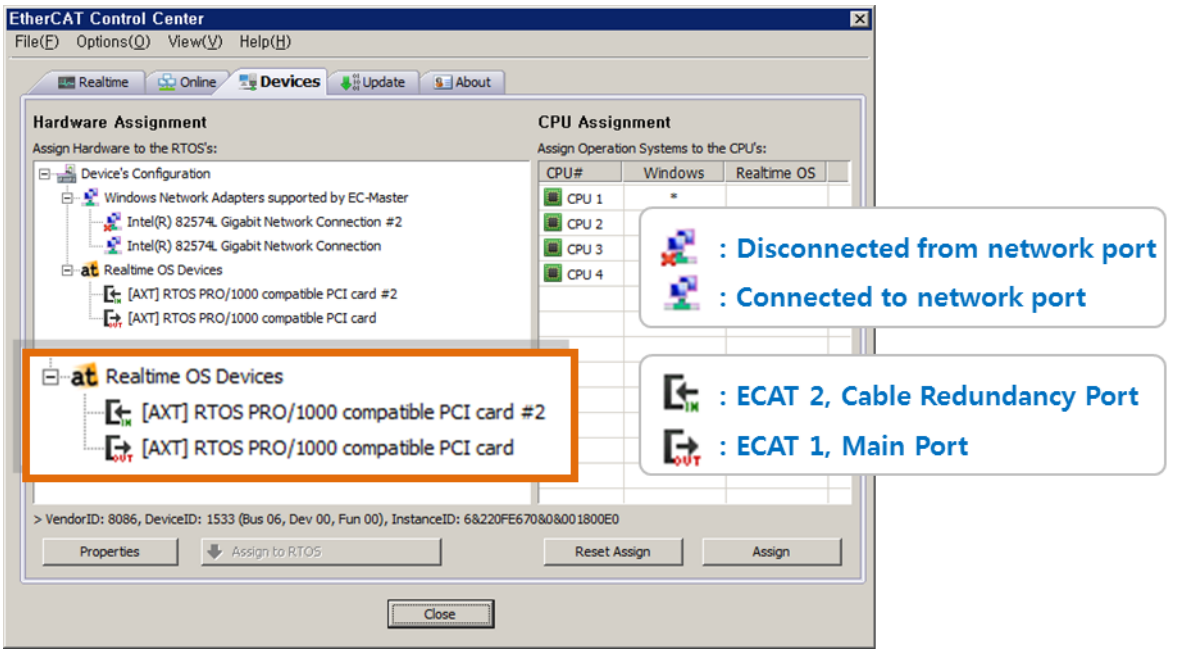


4. Select the network port to register in the RTOS then the "Assign to RTOS" button will be activated. If you click the button, a confirmation message window will appear. Click "OK" button to register the selected network port to RTOS.

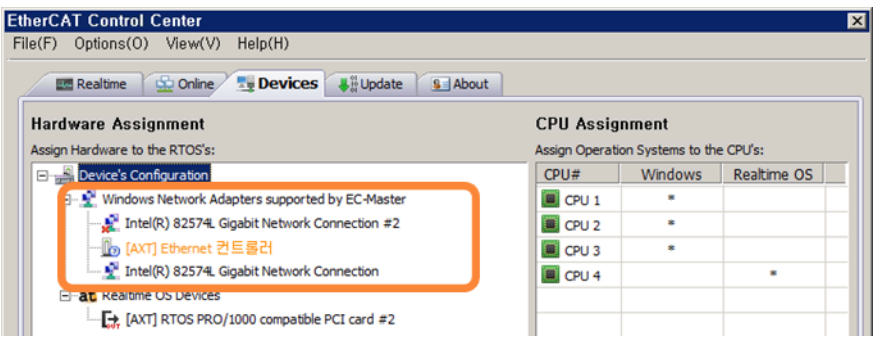


5. If the driver registration is successful, "Realtime OS Devices" item will be newly created in the Assign

hardware to RTOS's tree and the network port will be registered in "Sub" item. If you select the remaining [AXT] port and then click the "Assign to RTOS" button again, it is registered in the "Realtime OS Devices" item and is used as the EtherCAT Cable Redundancy port.



6. Among the network ports in the Hardware Assignment tree, the network port marked in orange is the port that the driver has problems. In this case, reboot the control system and register driver.



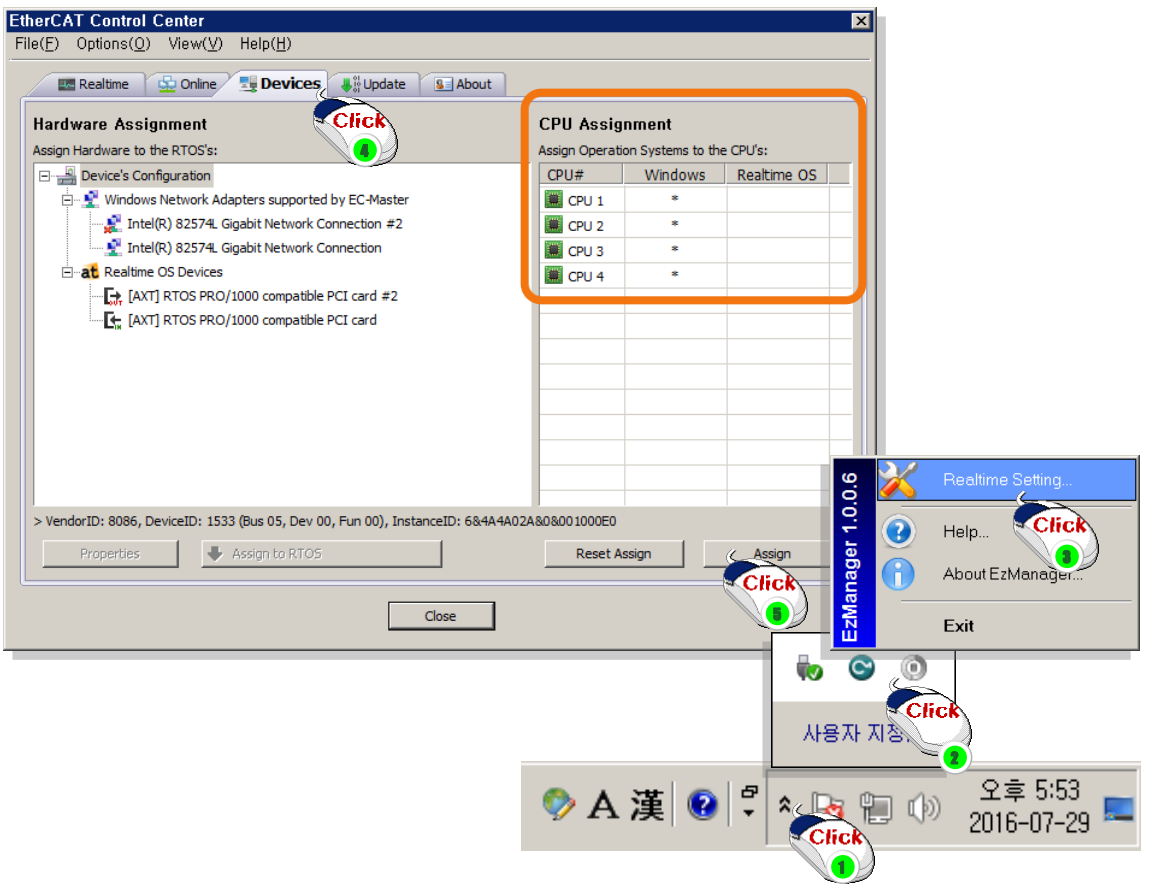
Caution) While the EtherCAT Master is running, the network port cannot be changed to RTOS or Windows.

2.2 Allocate CPU Core and Memory to RTOS

The EtherCAT Master provided by our company is a software type EtherCAT controller driven by Real Time Operating System (RTOS). Therefore, in order to run EtherCAT Master, it is necessary to allocate memory separately from the independent CPU core.

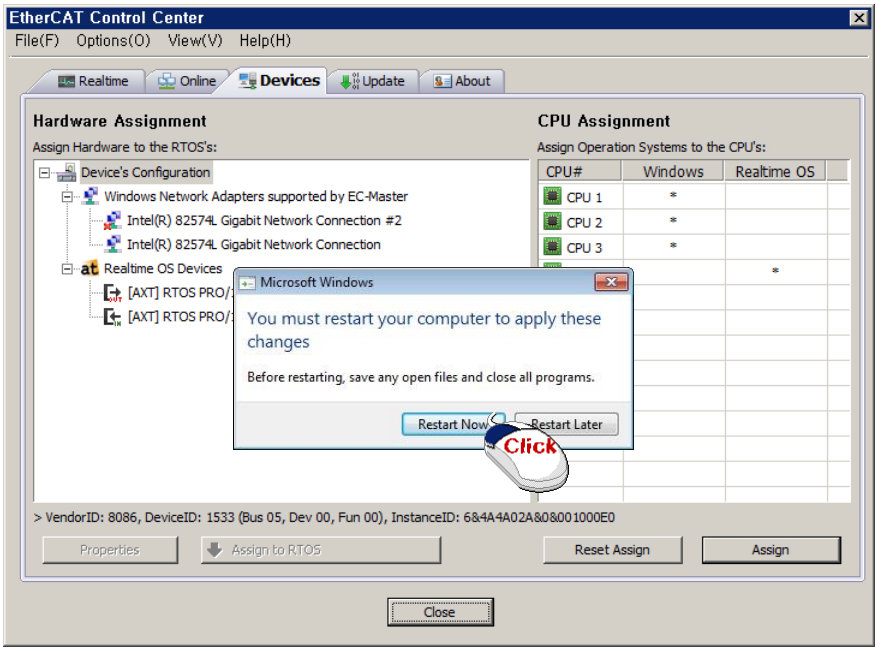
In general, all tasks that required to run the EtherCAT Master are done automatically when you install the EzSoftwareUC product. However, if problems occur during the installation process, you can manually allocate the CPU core and memory as shown below.

1. Click the EzManager tray icon in the bottom right corner of the Windows task bar and select "Realtime Setting...". Select the "Devices" tab from the EtherCAT Control Center to allocate the CPU core and memory.



2. All four CPU cores are assigned to the Windows item in the "CPU Assignment" list. Click the "Assign" button to allocate the CPU core and memory required for the RTOS.

3. In order to operate the EtherCAT Master properly, it is necessary to reboot the system after allocating the CPU core and memory. Click the "Assign" button then the message box about system rebooting appears. Click "Restart Now". All the tasks are completed after reboot the system.



3. EzManager

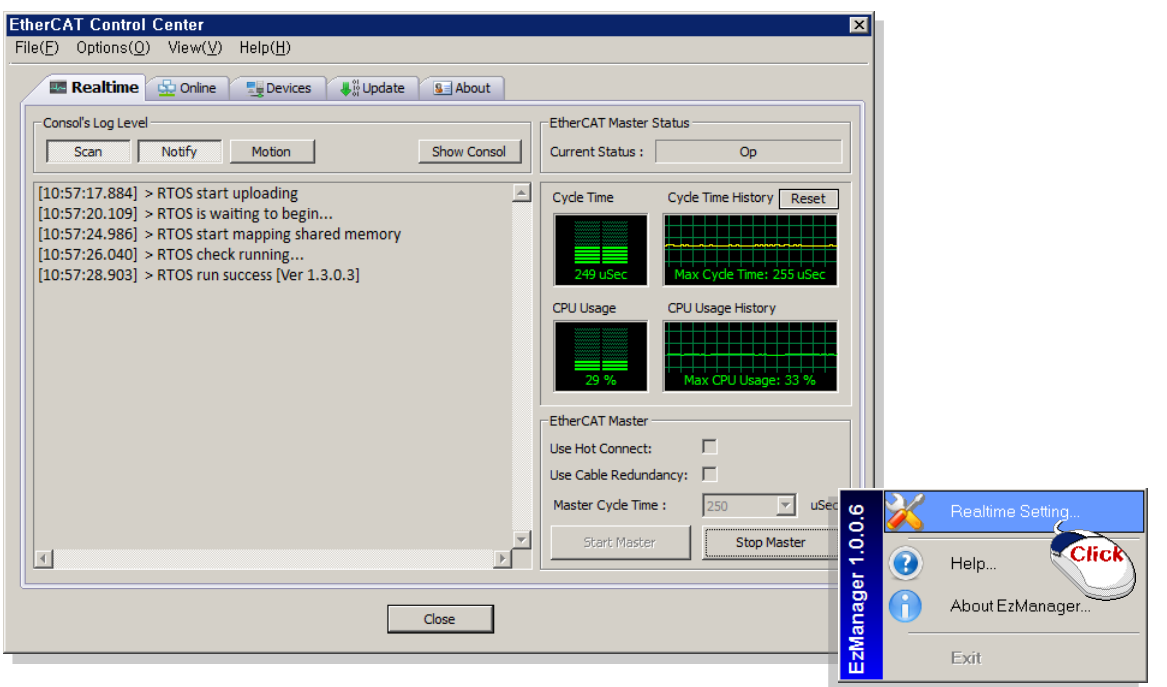
3.1 Layout

3.1.1 Tray Menu

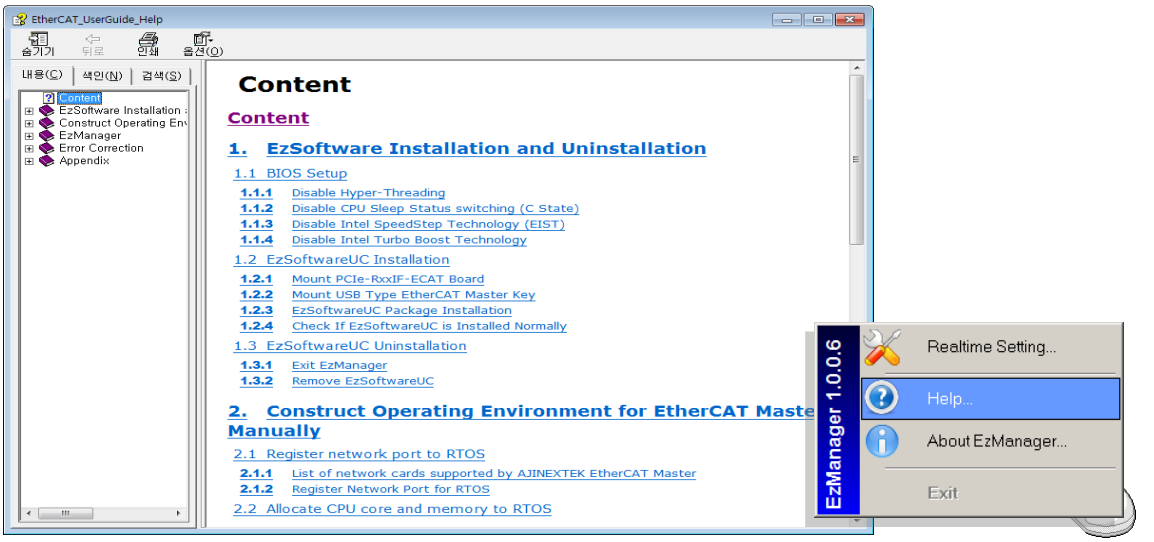
1. Tray menu appears when click the EzManager program icon at the bottom right of the Windows task bar. While the EhterCAT Master is running, the "Exit" button is disabled and cannot exit the EzManager program.



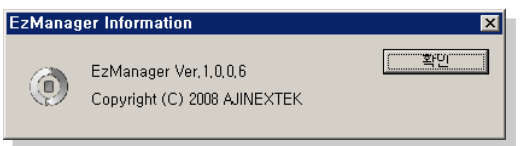
2. Select the "Realtime Setting ..." menu to run the "EtherCAT Control Center" to manage the EtherCAT Master. This Control Center can manage the EtherCAT Master, diagnose the slave status, and update files".

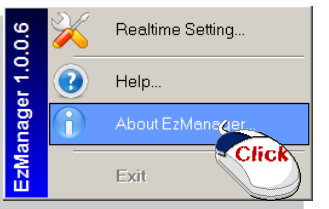


3. Select the "Help ..." menu to launch the EtherCAT User's Guide.



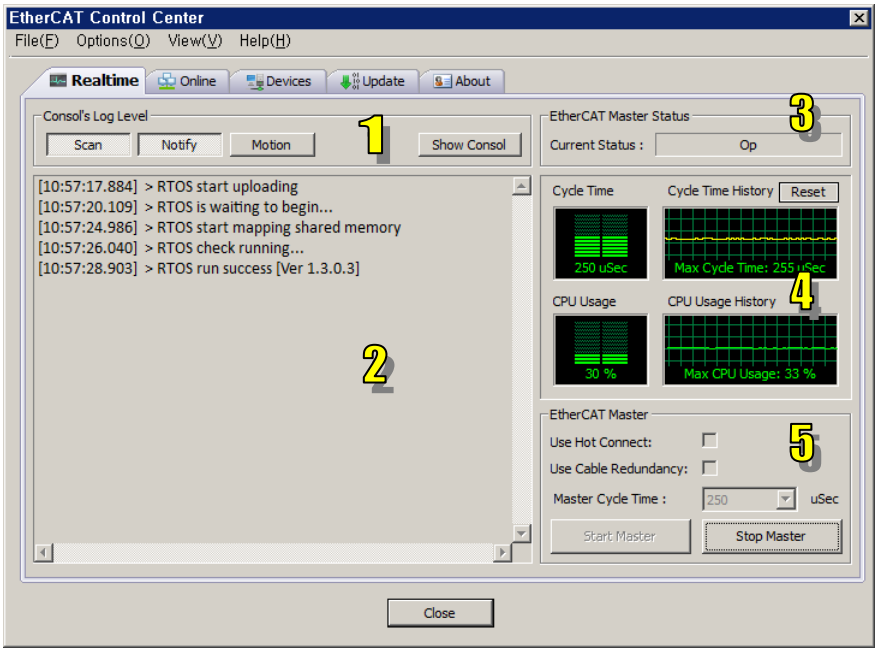
4. Select the "About EzManager ..." menu and the EzManager Information dialog box appears. You can check the EzManager version information.



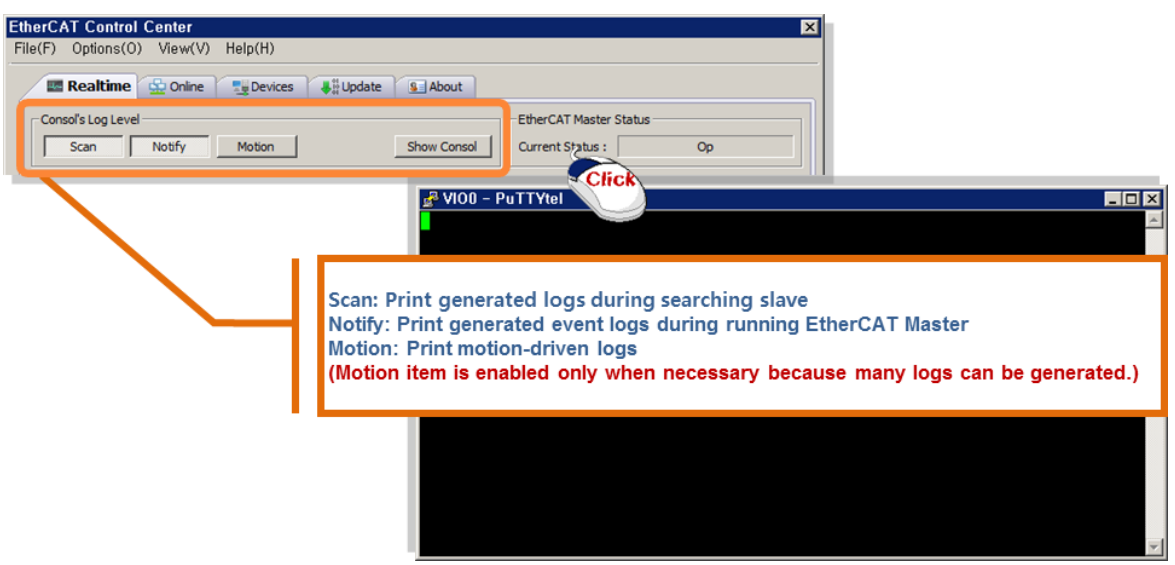


3.1.2 Realtime Layout

In the Realtime tab, you can start or stop the EtherCAT Master and check the running status of the RTOS on which the Master is running.

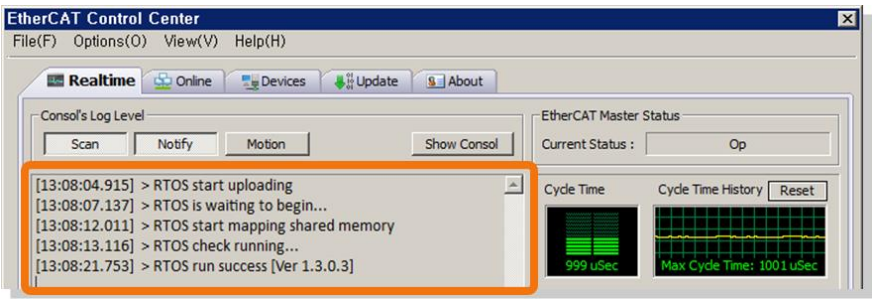


1. In the "Realtime" tab, you can check the logs that generated when the EtherCAT Master runs in the RTOS. Click the "Show Consol" button to open the logs console window.

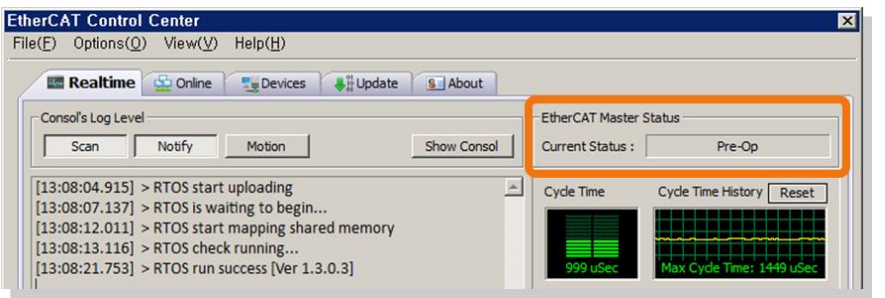


2. When the EtherCAT Master starts to run, the "RTOS start uploading" log is output, and the memory

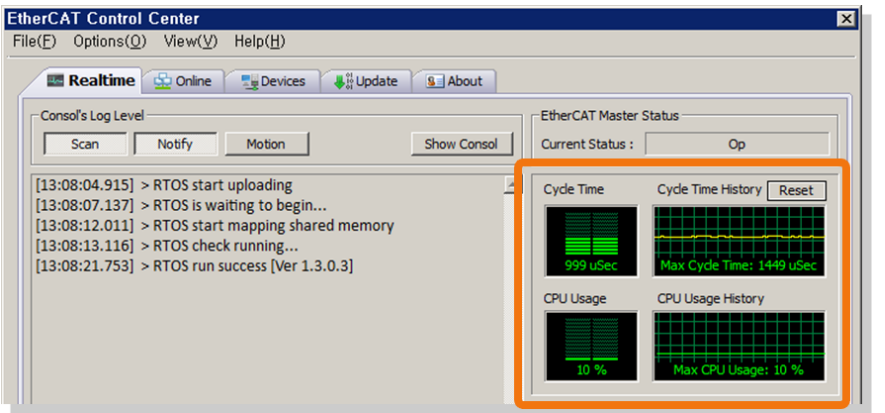
mapping, RTOS operation check, and the last success status with the version information of RTOS are output.



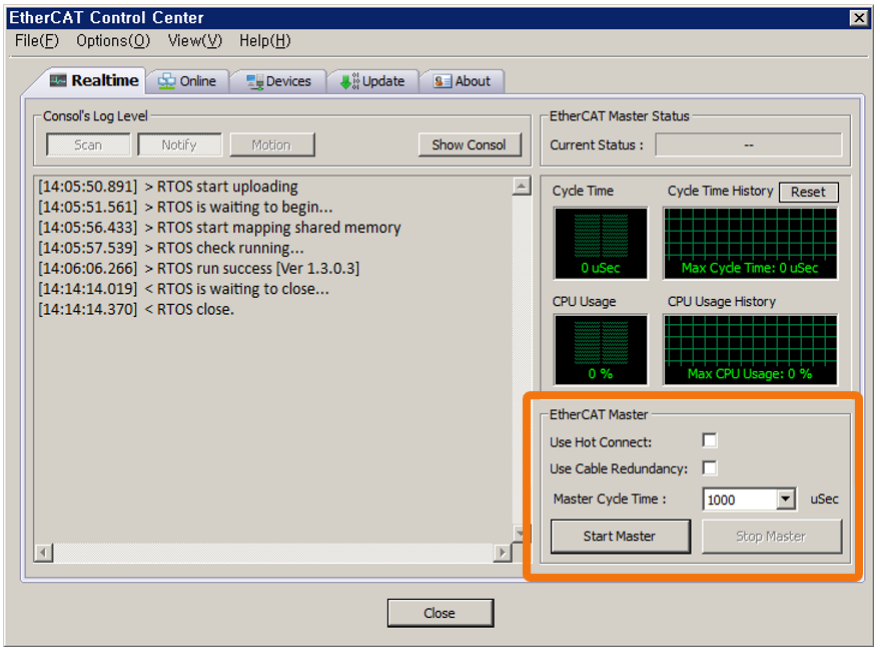
3. Output the status of EtherCAT Master. After the EtherCAT master is started, if all slaves connected to the network become controllable, it changes from "Pre-OP" to "Op" status.



4. Shows graphs of current CPU usage and cycle time of RTOS (Real Time Operating System) running EtherCAT Master.



5. You can start and stop the EtherCAT Master. Before starting Master, you can select to use Hot Connect and Cable Redundancy, and can change Cycle Time. All settings cannot be changed while the EtherCAT Master is running.

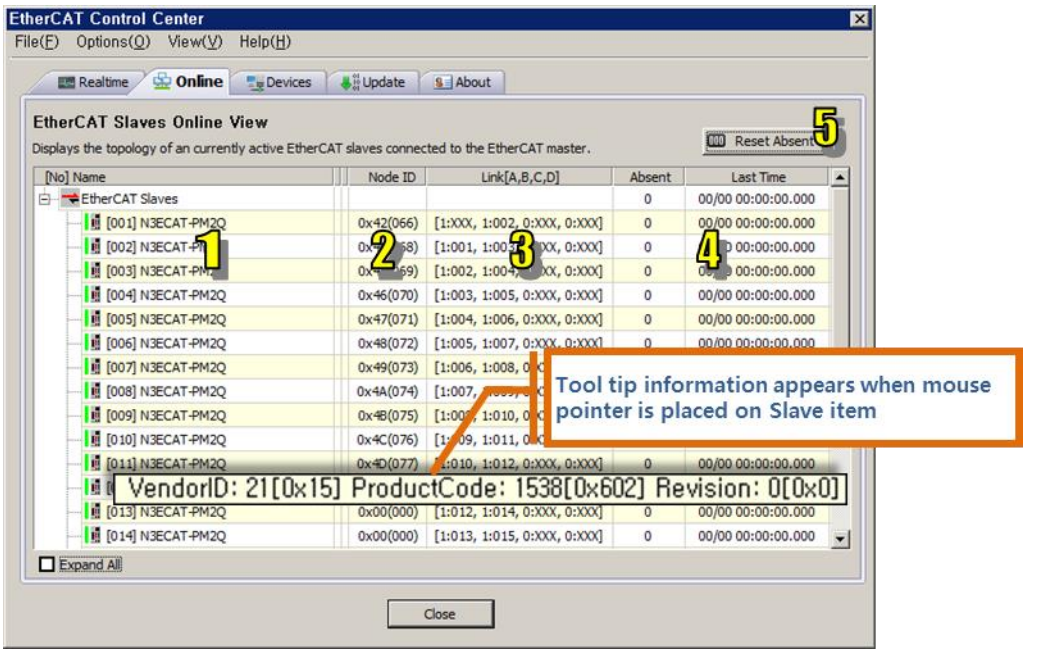


[Description of EtherCAT feature]

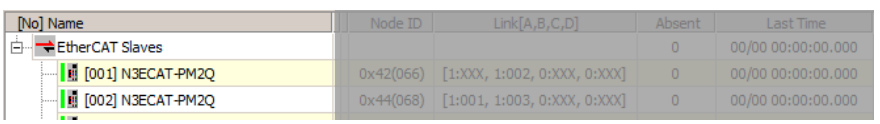
Feature	Description
Hot Connect	Ability to continue to operate Slaves normally connected to the EtherCAT Master when more than one slave in the network configuration is disconnected from the power or network connection. (If you change the use of Hot Connect, you must rescanning EzConfig program.)
Cable Redundancy	Using two network ports registered in the EtherCAT Master, any single broken connection in the network will not cause any slave to be disconnected

3.1.3 Online Layout

Online tab shows connection status of Slave connected to EtherCAT Master. You can use this information to check the cable contact and connection.



1. Check the network cable connection after connect the slaves to the EtherCAT Master at first. The first column "[No] Name" is the slave searched order and product name.



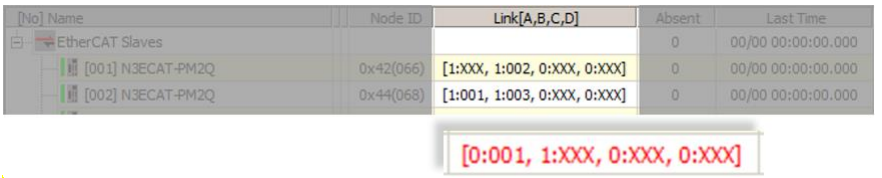


➢ Each icon represents the current state of the Slave product.

2. The second column "Node ID" is the slave's node ID. In general, the Node ID is the setting value of the rotary switch in each slave product.



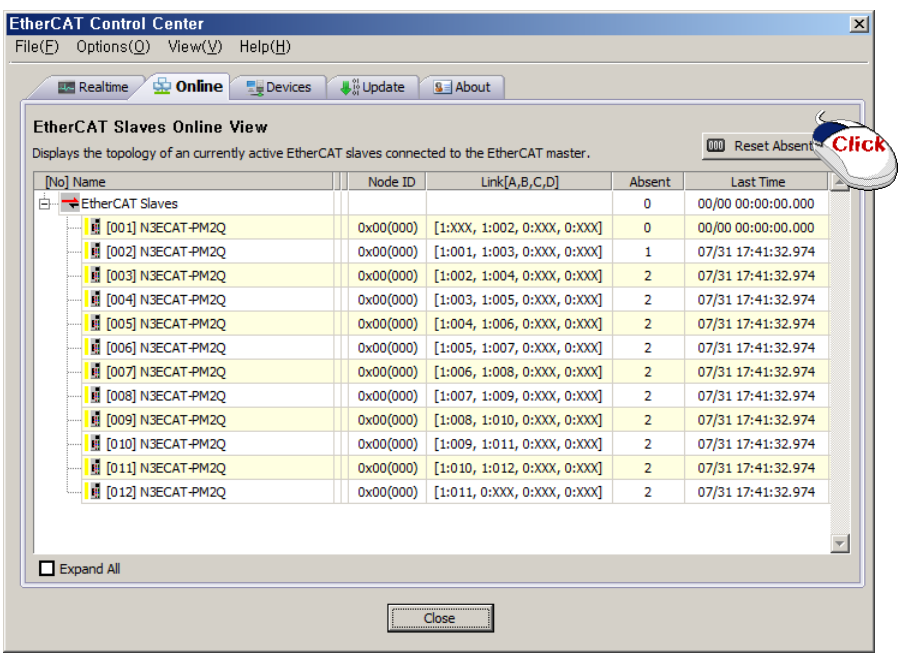
3. The third column "Link [A, B, C, D]" is the link status of each slave and slave number connected to the network port. Using this information, you can check the connection that made from the EtherCAT Master to the input port of the first slave and from output port of the first slave to input port of the next slave. If the cable connection is incorrect, the slave link information is displayed in red.



4. If EtherCAT Master is disconnected due to cable contact problem during normal operation, the "Absent" count value will increase. Even if the network connection is disconnected and reconnected in a moment, the EtherCAT Master detects it and displays the "Absent" count value and the last occurred time.

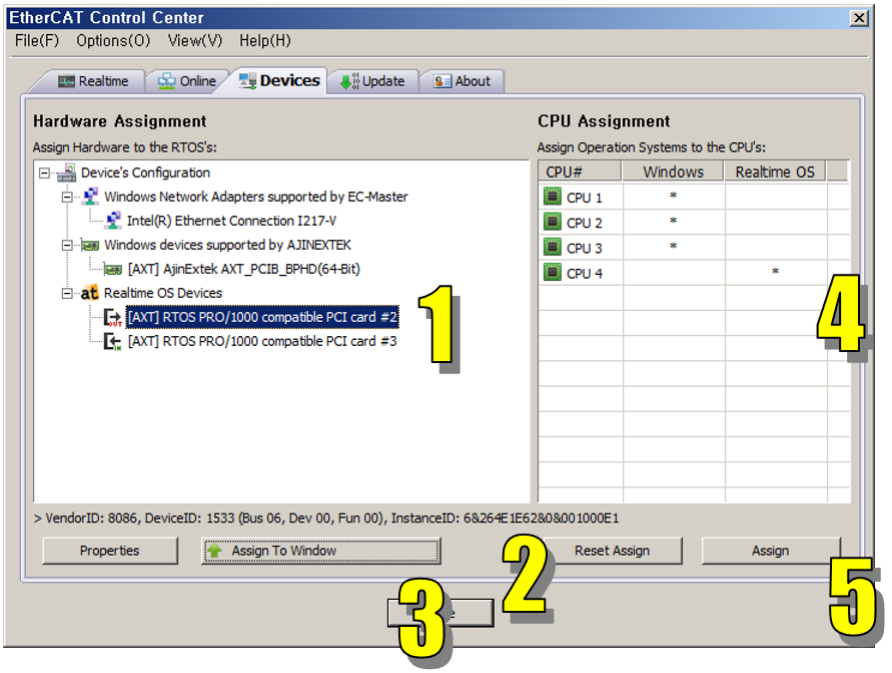


5. You can reset the recorded network breakdown information of each slave by pressing the "Reset Absent" button.



3.1.4 Devices Layout

The Device tab shows the hardware resources of the control system required to run the EtherCAT Master and can be specified for use with the EtherCAT Master.



1. You can register device drivers of network ports in Hardware Assignment tree from Window to RTOS or from RTOS to Window. All settings are possible only when the EtherCAT Master is stopped.

2. Select the device to be changed in the Hardware Assignment tree and the device information will be output. The output information includes the vender ID, device ID, and bus number that can identify the hardware device

```
> VendorID: 8086, DeviceID: 1533 (Bus 06, Dev 00, Fun 00), InstanceID: 6&264E1E62&0&001000E1
```

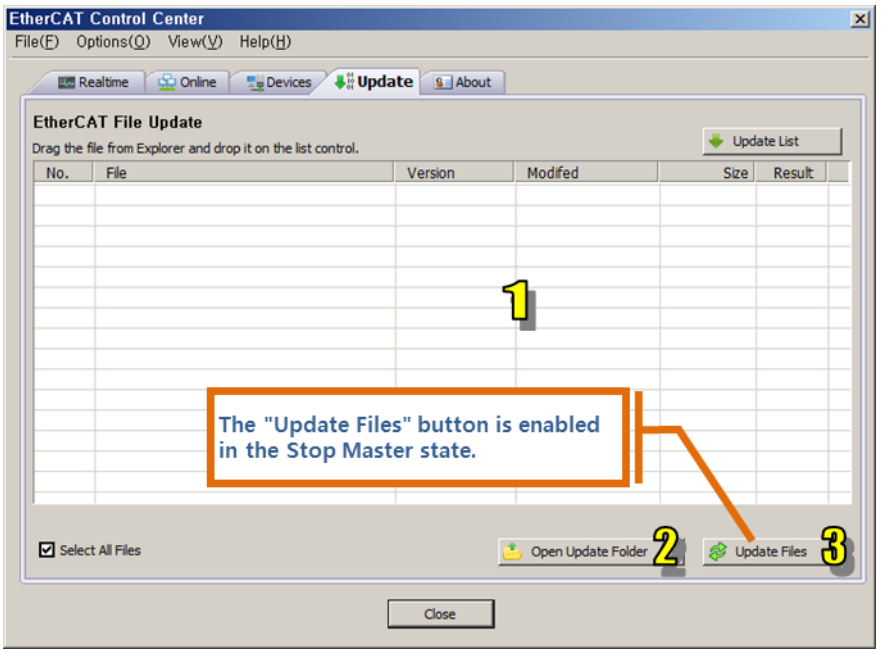
3. Select the device to be changed in the Hardware Assignment tree then the "Assign To RTOS" or "Assign To Window" button will be activated according to the situation. When you click the button, driver can be changed after confirmation.

4. The CPU Assignment list shows the valid CPU cores available in the control system. One CPU core must allocate to the "Realtime OS" to operate EtherCAT Master normally.

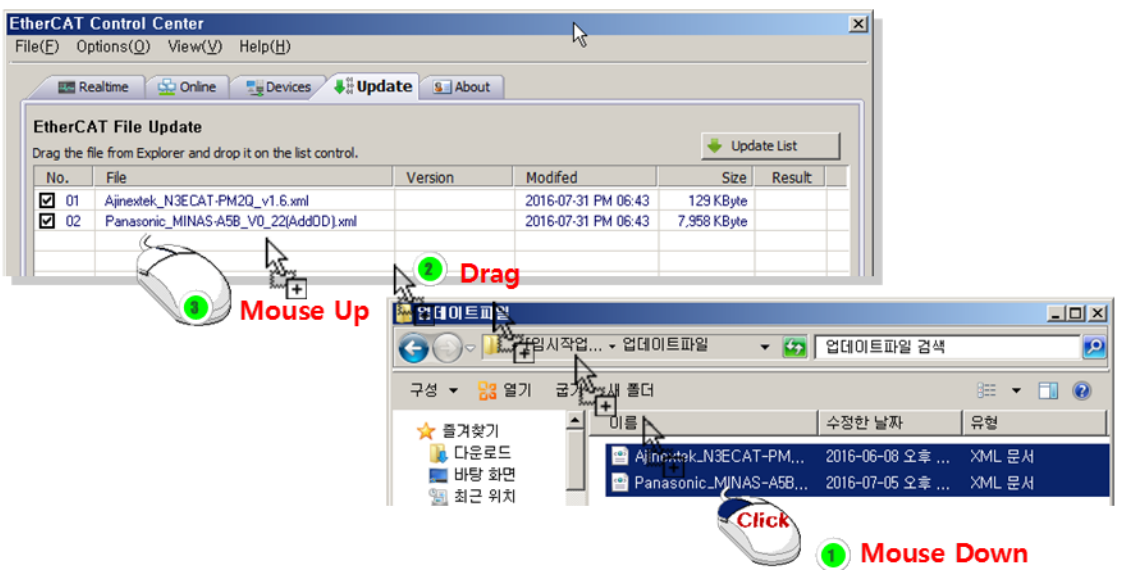
5. When the EtherCAT Master is stopped, the "Assign" or "Reset Assign" button is activated. Pressing the "Assign" button will allocate one CPU core and memory to the RTOS. Pressing the "Reset Assign" button will return the CPU core and memory allocated to the RTOS to the window.

3.1.5 Update Layout

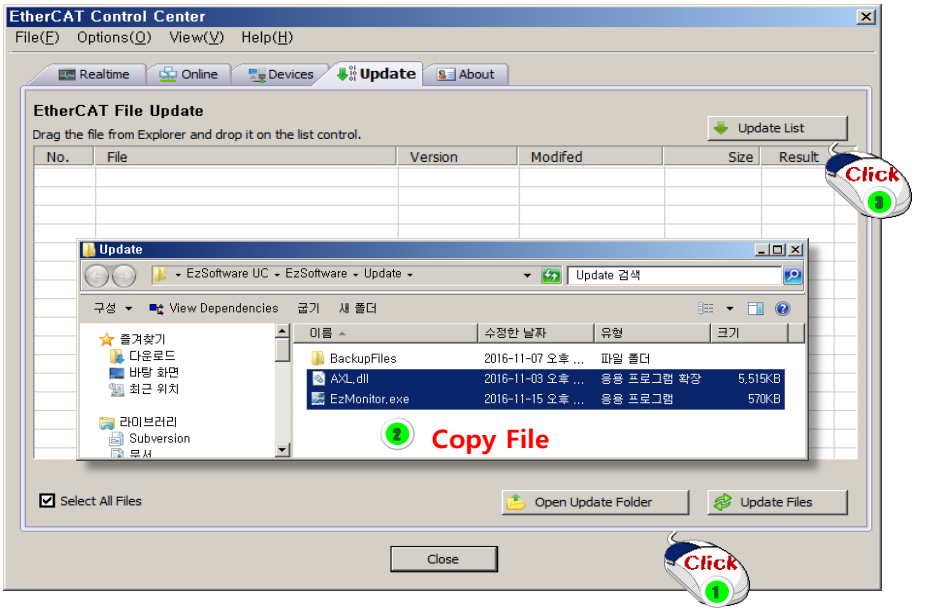
On the Update tab, you can update the necessary files for running the EtherCAT Master. To proceed update, copy the files to the update folder or to the update list by drag & drop. Update can be done while the Master is stopped, so click the "Stop Master" button on the Realtime tab to stop the Master



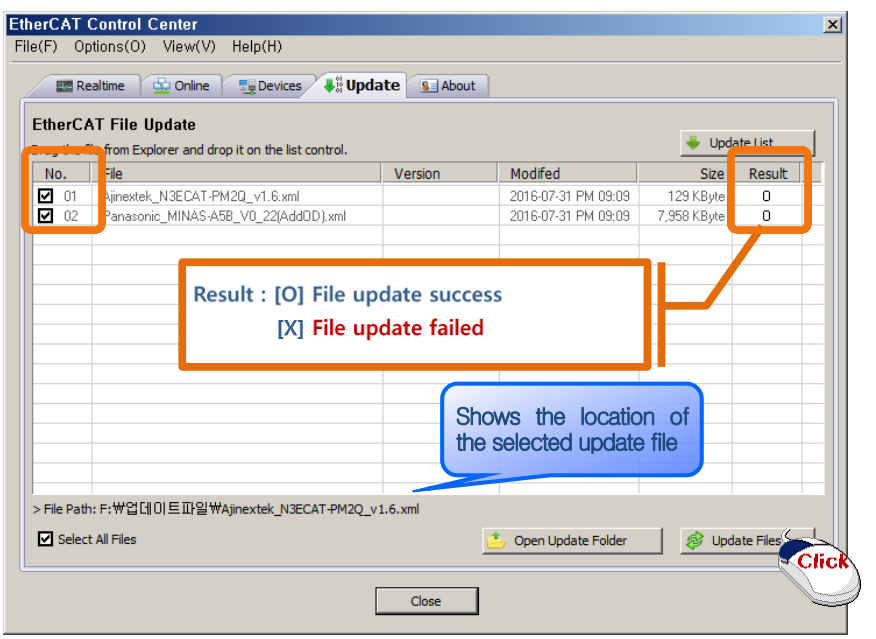
1. To register files and folders that needs to be updated on the update list, drag and drop from Windows Explorer to the update list.



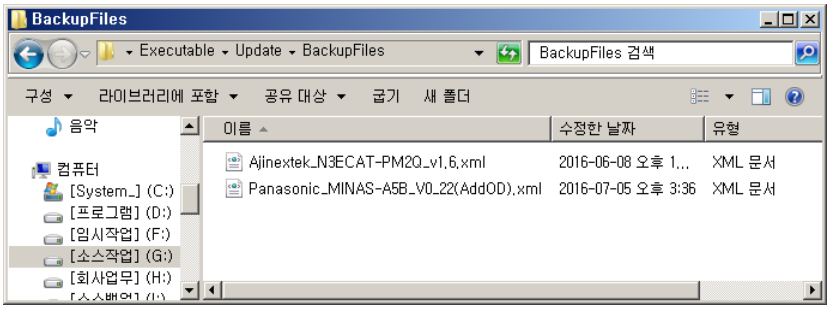
2. In a different way, copy the files to be updated to the explorer that appears by clicking the "Open Update Folder" button (Location: "EzSoftware UC installation folder> EzSoftware> Update"). After closing the update folder, click the "Update List" button and the files will be registered in the update list.



3. You can select and deselect the files to update in the update list. When you have finished selecting the update file, click the "Update Files" button to automatically update the files.



4. Once the update is complete, the files are automatically backed up to the "BackupFiles" folder in the "Update" folder. If the update is failed or you want to restore to an earlier version, you can use it.

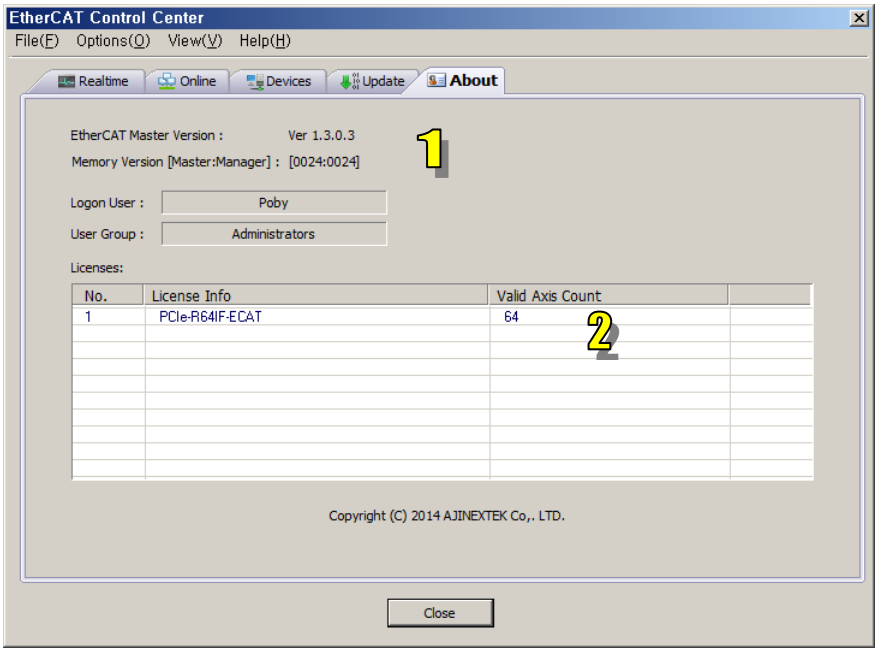


[List of files that can be updated]

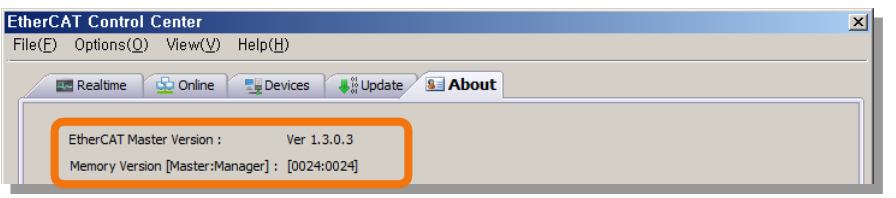
File	Description
EzConfig.exe	Config program to run product search and test program of functional modules
DIOAgent.exe	Program to test Digital Input/Output products
AIOAgent.exe	Program to test Analog Input/Outupt products
MotionAgent.exe	Program for Motion products to run and perform functional tests
CounterAgent.exe	Program to test Counter products
EzSpy.exe	Spy program showing AXL library called from other application
EzTerm/MR.exe	Program to read / write parameters, check alarm, input / output status of network products
EzUpdater.exe	Program to update firmware of network products
EzMonitor.exe	Program that display data graph (speed, positon, torque, contact status, etc.) generated by EtherCAT Master
EzManager.exe	Program to manage EtherCAT Master and diagnose network status
AXL.dll	Standard library to control the products provided by AJINEXTEK
EzBasicAxl.dll	Base library to access to the hardware provided by AJINEXTEK
HCI.dll	DLL used by EzConfig.exe
AXTECMaster.bin	EtherCAT Master Binary file
AXTECDeviceDataBase.xml	Information file of Slaves supported by EtherCAT Master
AXTECAlarmList.xml	Alarm information file of EtherCAT driver products
ESI File	Information file containing information of EtherCAT Slaves

3.1.6 About Layout

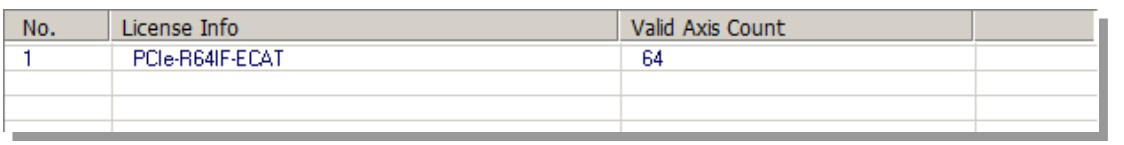
In the About tab, you can see version information of important files related to EtherCAT Master and license information



1. When the EtherCAT Master starts up, you can check the version information of the EtherCAT Master and the memory version used for data exchange between the Windows application and the EtherCAT Master. If this memory version does not match, you will not be able to control EtherCAT controller, so you must match the version of the EzManager program with the version of the EtherCAT Master file (AXTECMaster.bin).

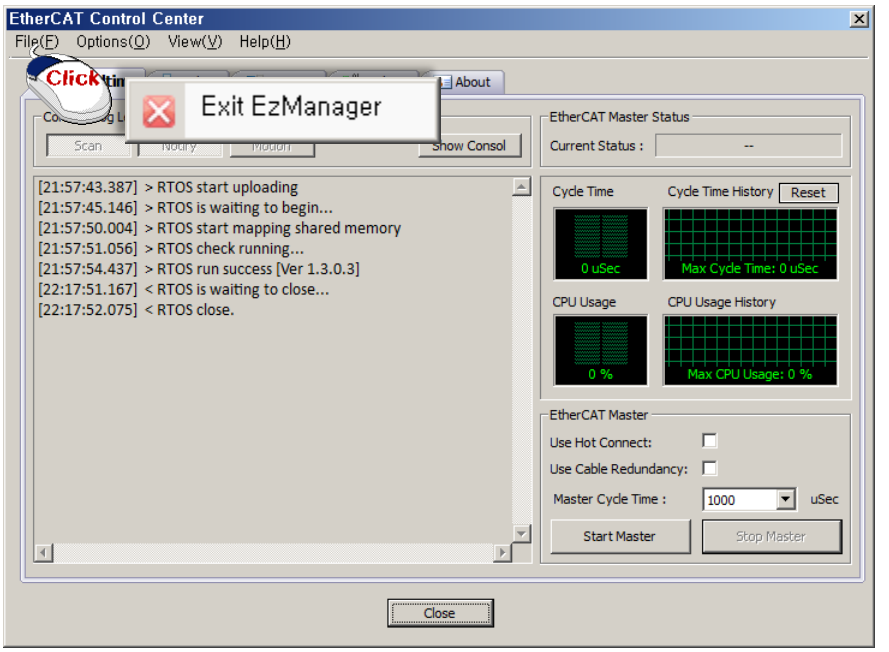


2. The EtherCAT Master can only be run on system with valid licenses. You can find issued licenses for the current system in the "Licenses" list. If there is "PCIe_RxxIF-ECAT" in the license list, it means you can run the EtherCAT Master and can control the number of axes in the "Valid Axis Count" item.



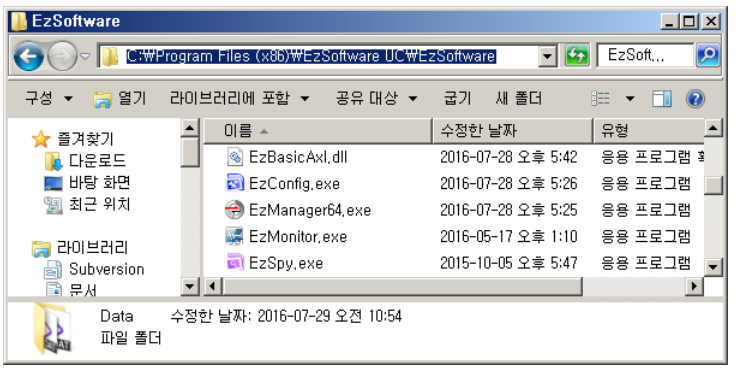
3.2 Menu

3.2.1 File Menu



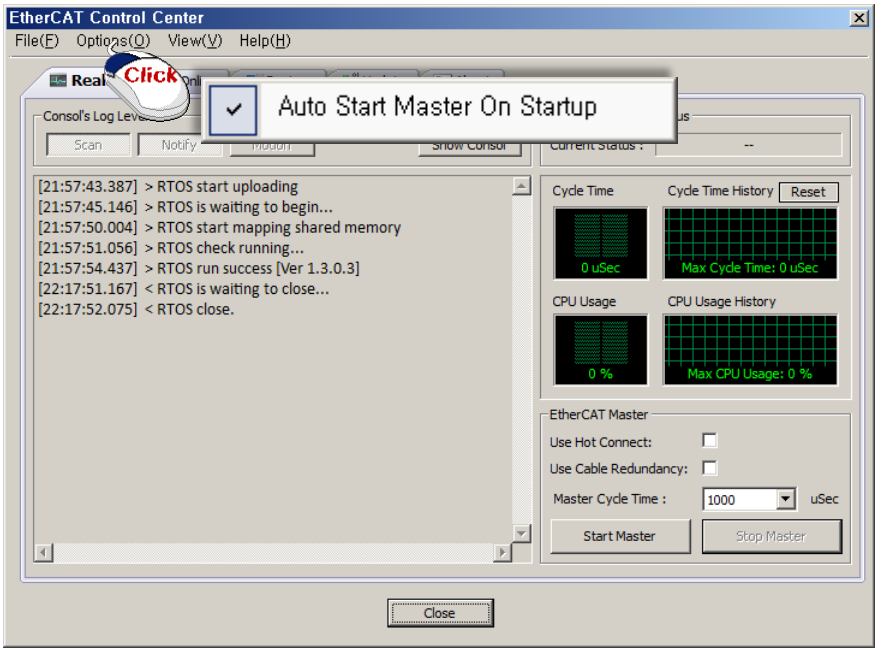
1. When the EtherCAT Master is stopped, you can exit the EzManager using the "Exit EzManager" menu. If EzManager is terminated, it cannot be used as an EtherCAT controller, so you should only use it if you remove the package or if you have a problem with the EtherCAT Master.

2. EzManager is automatically restarted after rebooting the system after exiting EzManager. If you want to run EzManager without rebooting, "EZManagerXX.exe" program is located in "EzSoftware" folder where EzSoftwareUC is installed.



3.2.2 Option Menu

EzManager is registered in the startup program, so it starts automatically when Windows starts. When EzManager starts, EtherCAT Master will be started automatically by default.

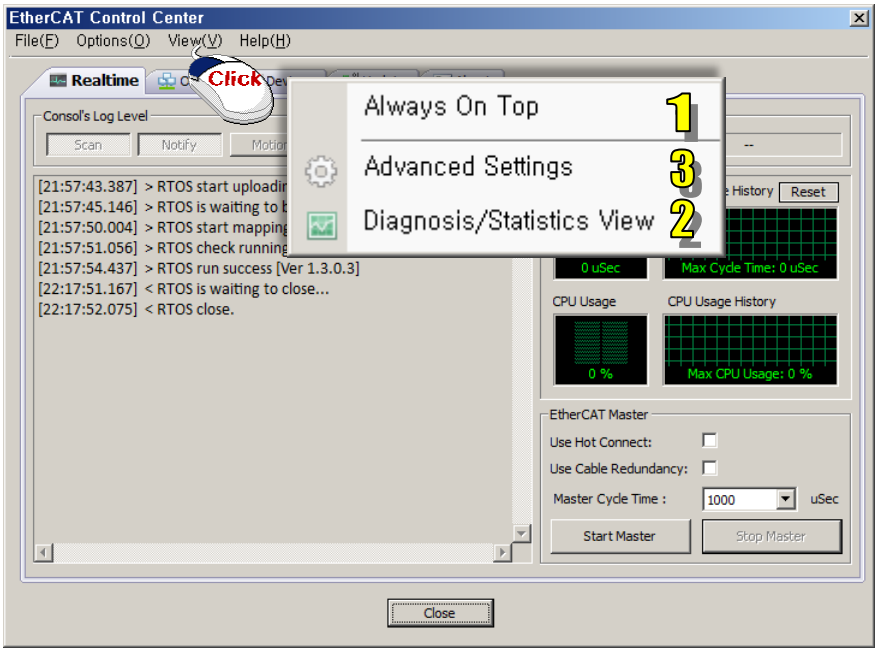


1. You can prevent the EtherCAT Master from starting automatically when EzManager is executed by using "Auto Start Master On Startup" menu in Option menu. If unchecked, EzManager will not start the EtherCAT Master automatically from the next execution.

2. This setting is maintained after rebooting the system, so you must manually start the EtherCAT Master after unchecking it. If the AXL library is used without the EtherCAT Master being started, the EtherCAT Master is not considered to be in the system, so you can't control EtherCAT Slaves.

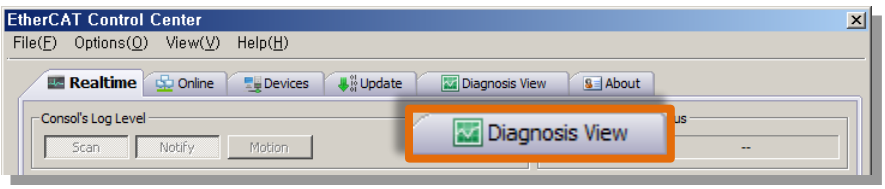
3.2.3 View Menu

You can check the performance of the control system by analyzing the operation status of EtherCAT Master through "Diagnosis / Statistics View" menu.

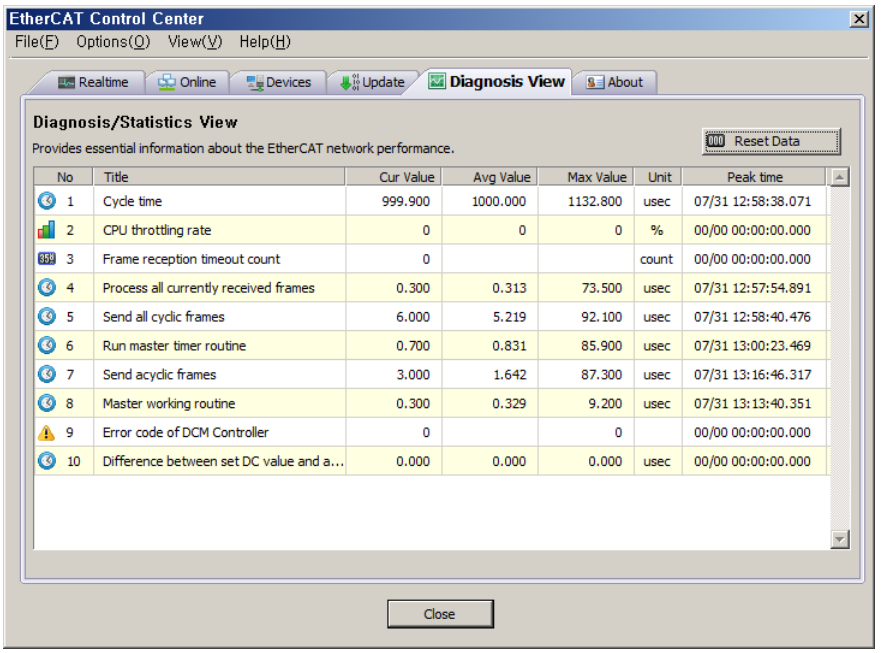


1. "Always On Top" menu allows you to place the EtherCAT Control Center always on top. With this menu, the screen does not disappear even if the focus of the mouse is transferred to another application program. This setting is maintained after rebooting the system.

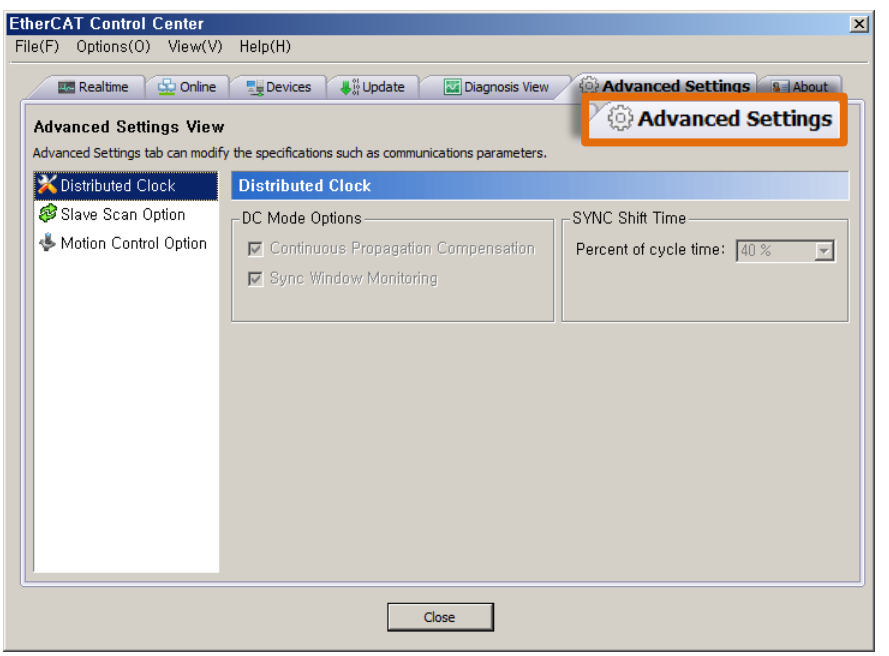
2. "Diagnosis / Statistics View" menu creates a new "Diagnosis View" tab. You can check and analyze the operation status of EtherCAT Master by using this tab.



3. The "Diagnosis View" tab allows you to check the current time, average time, and maximum time used inside the EtherCAT Master. "Peak time" means the system time when the maximum period is updated.

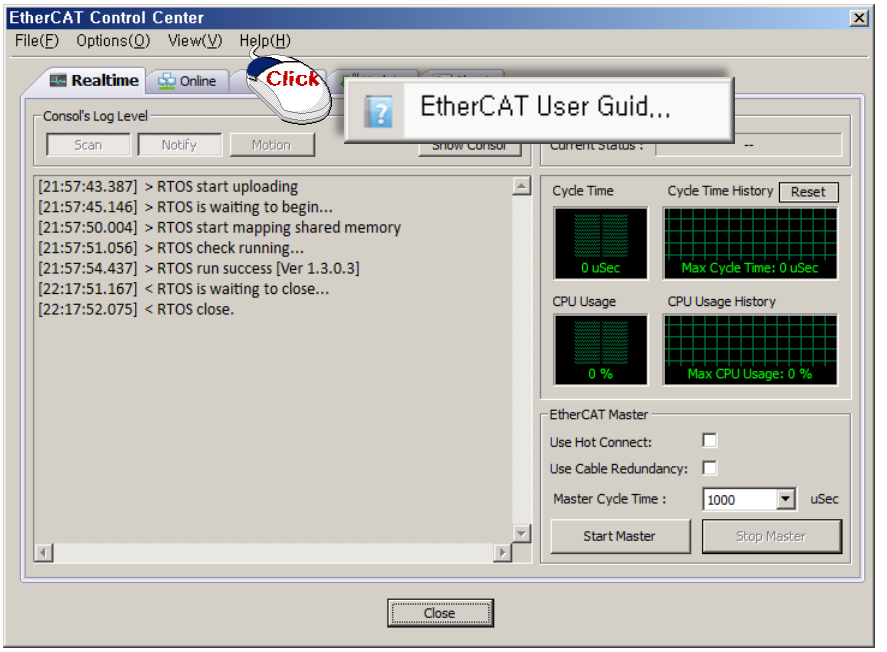


4. "Advanced Settings" menu creates the Advanced Settings tab. You can change the DC option, SYNC Shift Time, and Slave search options of the EtherCAT Master using this tab. It can be changed only when the EtherCAT Master is stopped and it will be closed automatically when the EtherCAT Master is restarted.

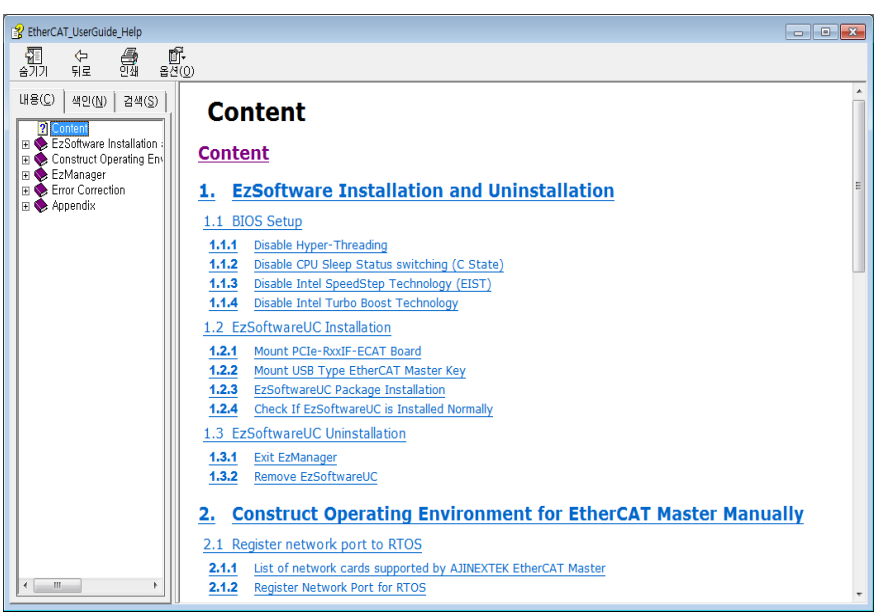


3.2.4 Help Menu

You can run the EtherCAT User Guide directly from the Help menu. The EtherCAT User Guide can be viewed immediately without a separate Viewer program.



1. Select "EtherCAT User Guide..." Menu. "EtherCAT_UserGuide_Help.chm" file located in "EzSoftware UC₩EzSoftware₩Helps" folder is displayed.

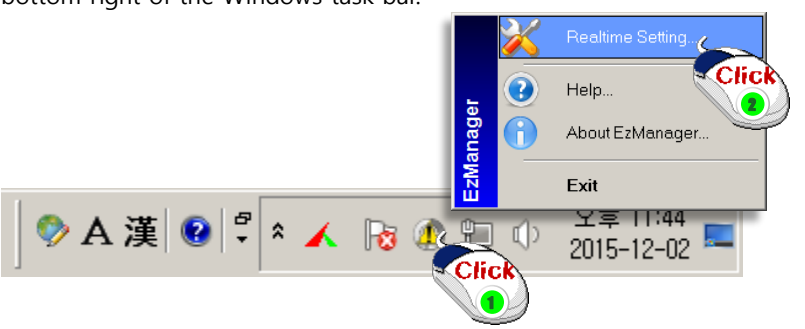


4. Error Correction

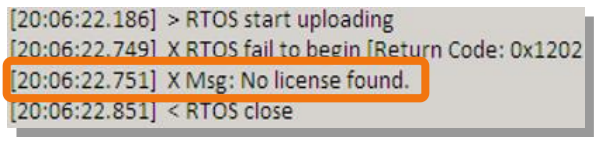
4.1 If EtherCAT Master is not running

4.1.1 If EtherCAT Master Key is not installed

1. After installing the EzSoftwareUC package for EtherCAT and rebooting the system, the EzManager program will be executed automatically. EzManager is a program that runs and manages EtherCAT Master. To check whether EtherCAT Master is running with EzManager, click the EzManager tray icon at the bottom right of the Windows task bar.



2. In the "Realtime" settings tab, check the logs to see why the EtherCAT Master was not running. If the EtherCAT license is not recognized, you can check the message "No license found."

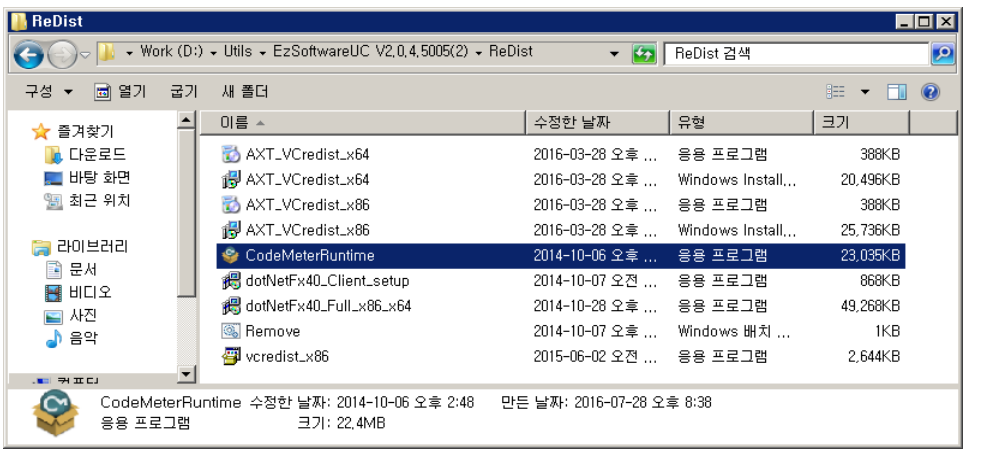


4.1.2 If License Manager Program (Codemeter) is not installed correctly

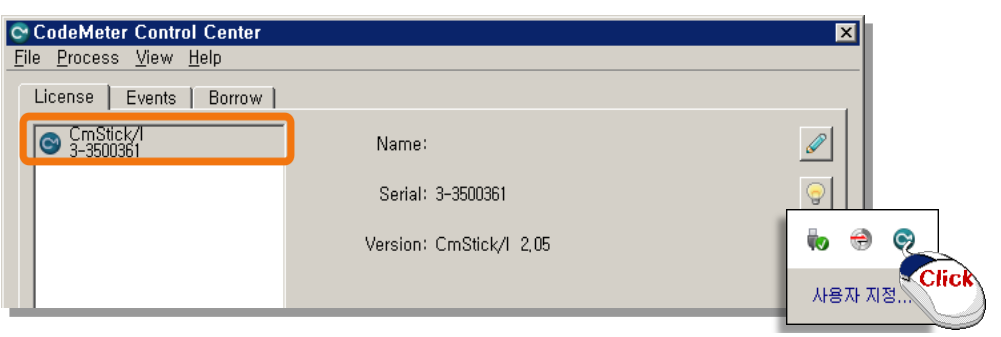
1. If the CodeMeter program, which is installed automatically when installing the EzSoftwareUC package, is not installed correctly, the license may not be recognized and errors may occur. Make sure the CodeMeter program is running at the bottom right of the Windows task bar where you see the Tray icon of the running program.



2. If CodeMeter program is not running, please install Codemeter manually using "CodemeterRuntime.exe" program in "ReDist" folder included in EzSoftwareUC installation package file.

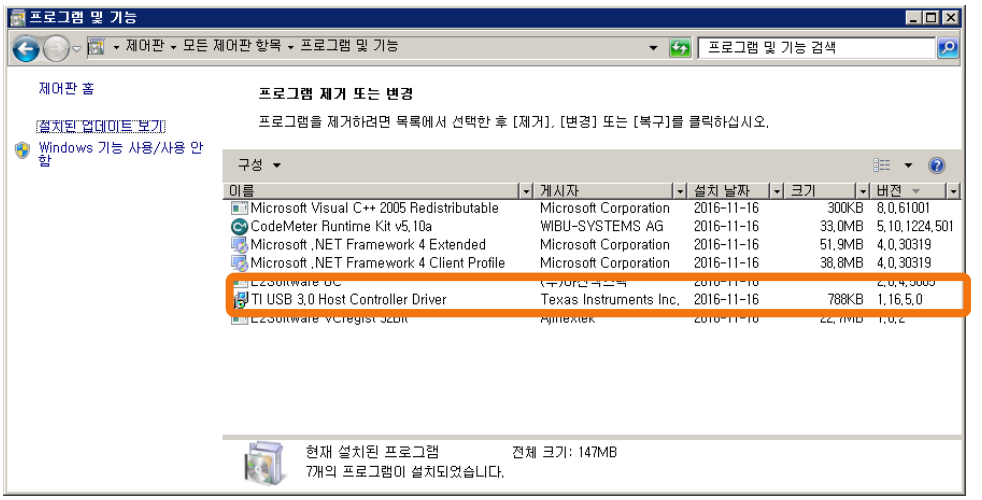


3. To check if the program is working normally click the icon of the running CodeMeter program then "CodeMeter Control Center" will appear. If there is more than one "CmStick" item in the left list of License tab, EtherCAT Master Key is working normally.



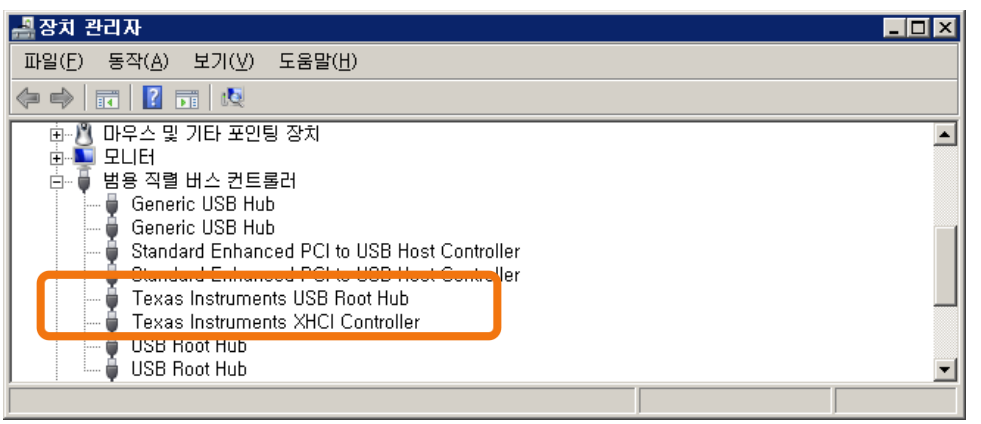
4.1.3 If EtherCAT Master Key Driver is not installed

1. When installing the EzSoftwareUC package, the device driver to recognize the EtherCAT Master Key is automatically installed. If the device driver is not properly installed, the license may not be recognized and an error may occur. Click the "Programs and Features" item in the Windows Control Panel and make sure that "TI USB 3.0 Host Controller Driver" is installed.



2. If the "TI USB 3.0 Host Controller Driver" is not installed, run the file "Texas_Instruments_xHCI_Driver_v1.16.5.0_Multilanguage_WHQL.exe" in the "C₩Program Data₩Ajinextek ₩EtherCAT₩Drivers₩Texas_Instruments_xHCI_Driver_v1.16.5.0_Multilanguage_WHQL" to install the device driver manually.

3. If the device driver for EtherCAT Master Key is installed and Master Key is recognized normally, "Texas Instruments USB Root Hub" and "Texas Instruments XHCI Controller" items are registered in the "Universal Serial Bus Controller" item in Device Manager.

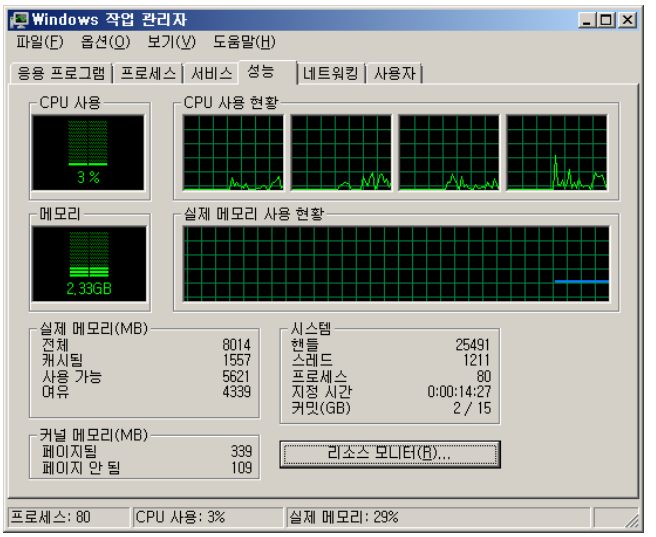


4.1.4 If EtherCAT Master Key is not recognized

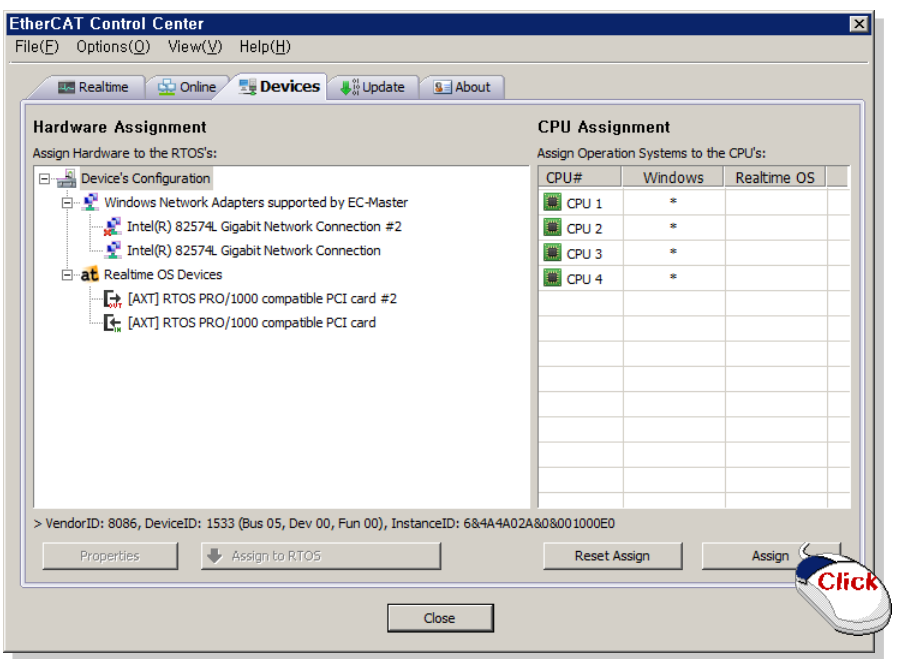
1. PCIe-RxxIF-ECAT board is basically equipped with USB EtherCAT Master Key. Even though the board is properly installed and the device driver for the codemeter program and master key is normally installed and running, the master key may not be physically recognized. If the issue is not resolved after re-installing the PCIe-RxxIF-ECAT card, please contact our Customer Support Team for technical support.

4.1.5 If CPU core for EtherCAT Master is not allocated

1. If the EzSoftwareUC package for EtherCAT is properly installed, one CPU core will be automatically assigned to the RTOS running EhterCAT Master. If the CPU core is not assigned, press the "Ctrl + Shite + ESC" key on the keyboard to launch Task Manager and select the "Performance" tab to identify the CPU core assigned to the window.

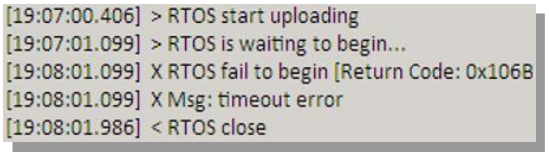


2. If the number of physical CPU cores is the same as the number of cores allocated to Windows, the CPU core is not allocated to the RTOS. Refer to "**2.2 Allocate CPU Core and Memory to RTOS**" to allocate CPU core and reboot the system.



4.1.6 If return code 0x106B is output and RTOS is not executed

If EzManager is executed automatically after EzSofrwareUC installed completely and reboot system, but the RTOS is closed with the following message (Return Code: 0x106B), the cause is as follows.

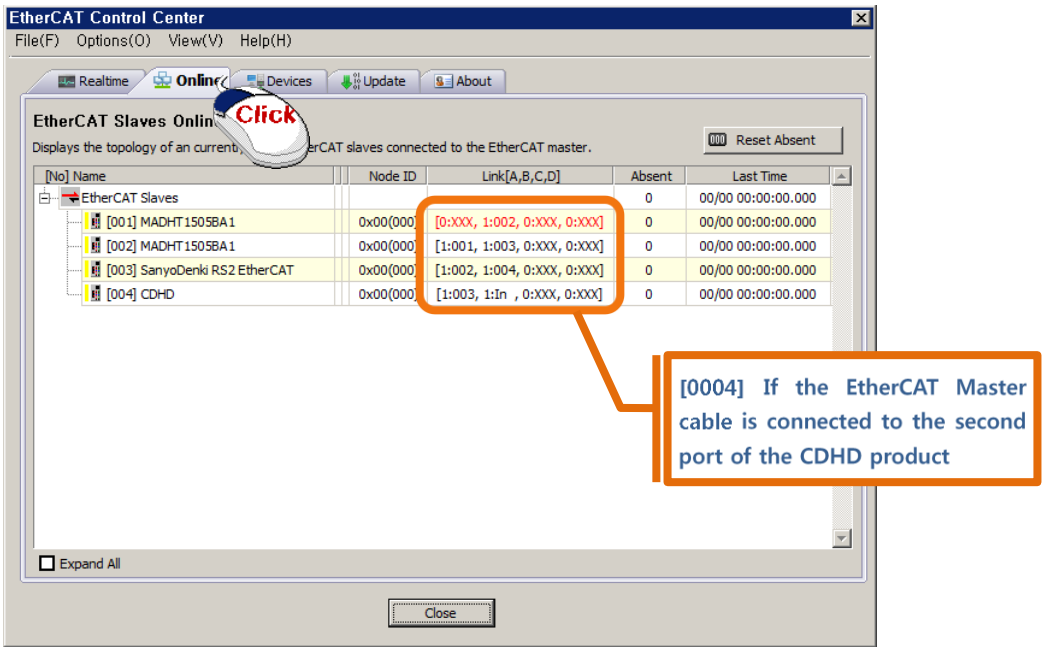


1. The EtherCAT Interface card is not installed so RTOS cannot be executed. Check the installation of the interface card.
2. The EtherCAT interface card is installed but the network adapter driver is not registered in the RTOS, so the RTOS cannot be executed. Please refer to "**2.1 Register Network Port to RTOS**" and register the network driver to RTOS.
3. The interface card is properly installed and the network adapter driver is also registered in the RTOS, but the memory for using the RTOS is not properly allocated. After the EzSoftware is installed, it may **need to reboot the system 2-3 times** in order for the memory to be allocated. Once the allocation is successful, it will be executed normally even after rebooting the system.

4.2 If EtherCAT Slave is not detected in EzConfig

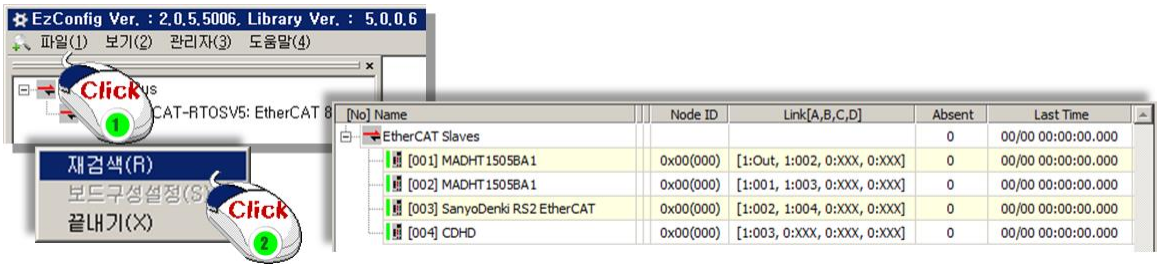
4.2.1 If Network Cable is not connected correctly

1. After connecting the first slave to ECAT1 of PCIe-RxxIF-ECAT board and running EzConfig UC program, all slaves connected to EtherCAT Master are detected and appear on the "Online" tab.



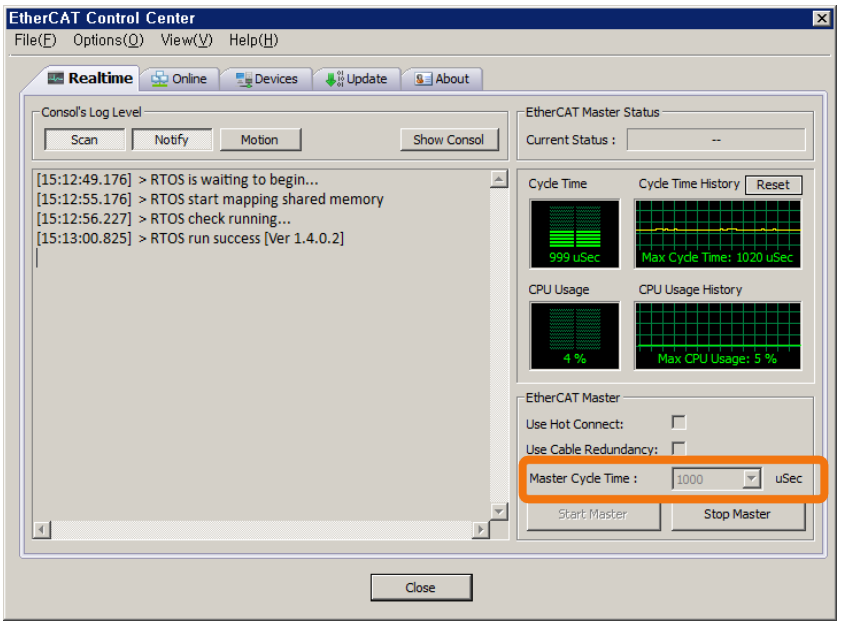
2. Link [A, B, C, D] of first slave is displayed in red to indicate a problem. Link A, B, C, and D indicate slave connection to the physical communication port and the number of the connected slave. Normally, it should be displayed as [1: Out, 1: 002, 0: XXX, 0: XXX]. [1: means that the first port of the slave is connected, and Out means that it is connected to the output port of the EtherCAT Master. 1: 002 means connected to the second slave. The remaining 0: XXX, 0: XXX] means that the third and fourth ports are not connected.

3. After reconnecting the cable, re-execute EzConfig or search the slaves using the "Rescan" menu in the File menu to check the cable is connected correctly.

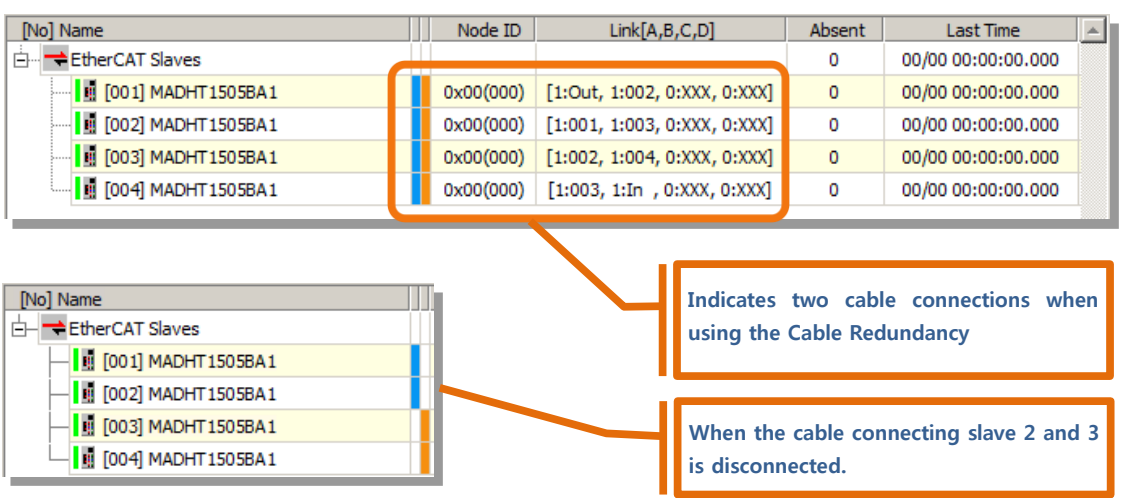


4.2.2 If slave is not detected

1. In order to use the Cable Redundancy function, if all cables are connected to the ECAT1 and ECAT2 ports on the PCIe-RxxIF-ECAT board (configured with Ling Topology), be sure to check the "Use Cable Redundancy". If it is not checked, slaves will not appear in EzManager's Online tab.

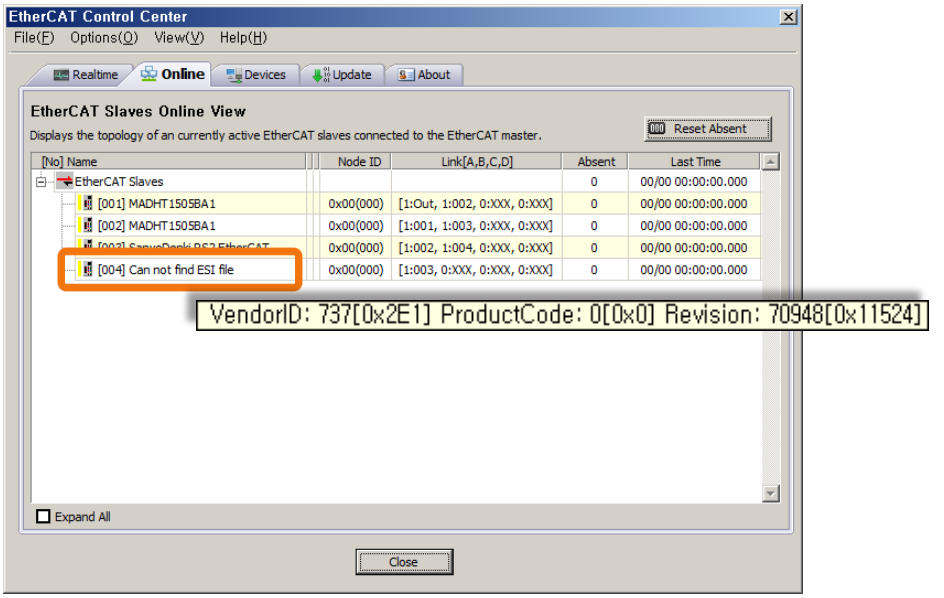


2. After closing all applications using the AXL library, click the "Stop Master" button on the "Realtime" tab to shut down the EtherCAT Master. Check the "Use Cable Redundancy" box and click "Start Master" button to restart the EtherCAT Master. After restarting EzConfig, you can check that slaves are searched normally.

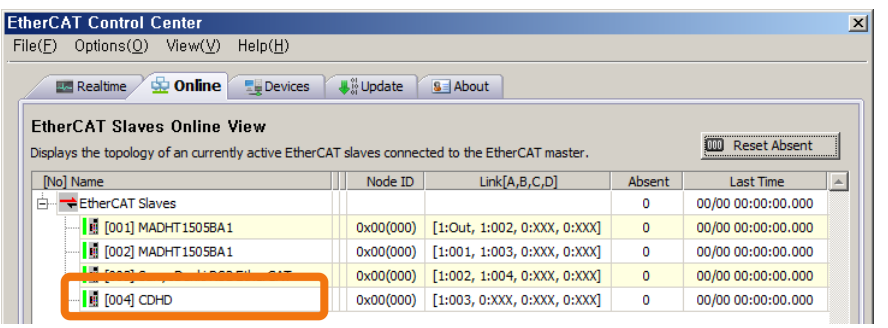


4.2.3 If ESI file does not exist for slave

1. The EtherCAT Master must have an ESI (EtherCAT Slave Information) file to control the Slave. If the product does not have an ESI file, it will be displayed as "Can not find ESI file" in the Online tab. Obtain the ESI file of the product corresponding to the VendorID, ProductCode and Revision number that appear when the mouse pointer is pointed to the module from the Slave supplier. Refer to "3.1.5 Update Layout" to proceed the update.



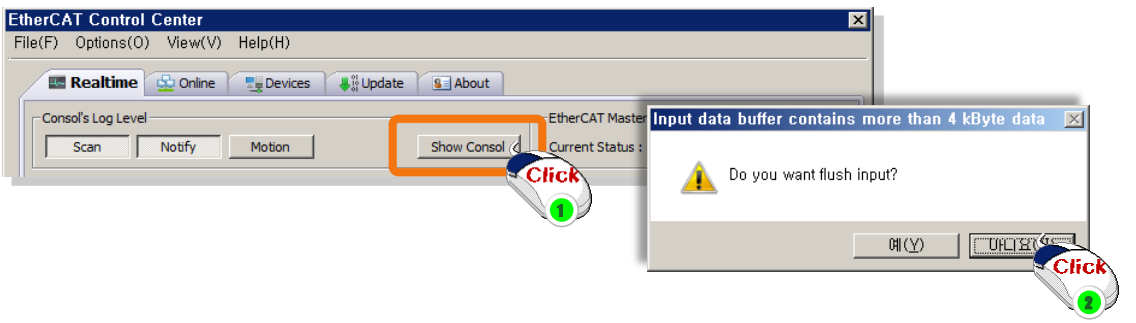
2. After updating the ESI file and restart EzManager, the name of the corresponding slave will be displayed on the Online tab and you can see that it is also found in EzConfig.



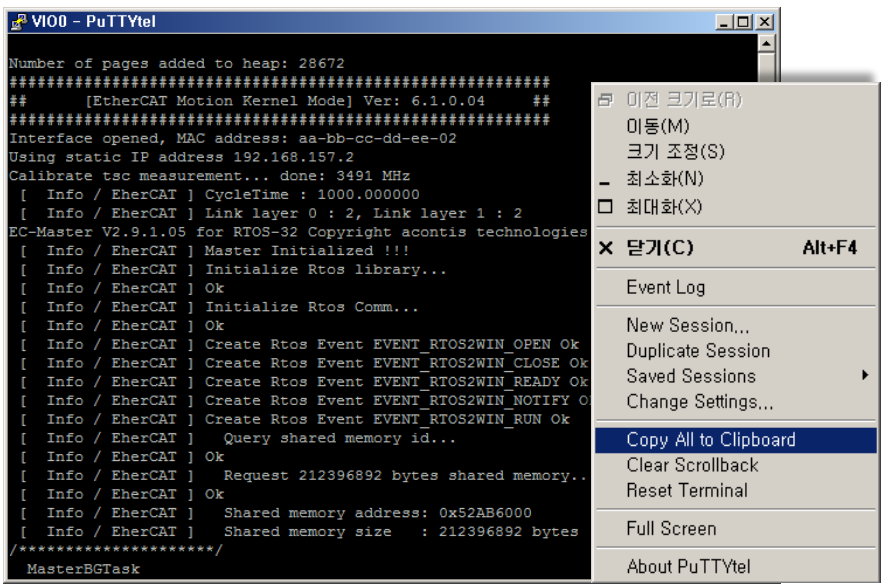
4.2.4 Save EtherCAT Master log

1. If the above solutions do not solve the problem, check the logs generated by the EtherCAT Master for more information.

2. In the "Realtime" tab, click the "Show Consol" button to open the logs console window. Select "No" in the dialog box asking whether to delete the message that occurred.



3. To launch the pop-up menu, right-click on the title bar of the console window to save the logs as a file. Select the "Copy All to Clipboard" menu will copy all the log in the console window to the clipboard.



4. Execute the Edit program (Notepad) and paste the logs (Ctrl + V). Please save the logs to a file and send it to our Customer Support team by e-mail. We can help you solve the problem through detailed cause analysis.

5. Appendix

5.1 Term Explanation

Term	Explanation
EzConfig	Configuration and operation support software tool of Ajinextek Co., Ltd. for each Motion, DIO, AIO, and Counter products installed in the control system
Real-Time OS(RTOS)	RTOS stands for Real Time Operating System and is an operating system developed for real-time applications. Designed focus on the CPU time management part of the operating system's functions. Real-time operating systems allow programmers control over process priority. The priority of the application may exceed the priority of the system program. The critical section of the system code is minimized, allowing the application's processing requests to be processed within a set time.
ESI	EtherCAT Slave Information File
Hot Connect	Ability to continue to operate Slaves normally connected to the EtherCAT Master when more than one slave in the network configuration is disconnected from the power or network connection.
Cable Redundancy	Using two network ports registered in the EtherCAT Master, any single broken connection in the network will not cause any slave to be disconnected